

Captured weight and boundary leakage bound the error of sample-based spectral functions

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Sample-based quantum diagonalization builds a subspace from a device’s measured bitstrings. From one protocol – computational-basis samples of a real-time evolved probe, no Hadamard test – we reconstruct the single-particle $A(\omega)$ and $A(k, \omega)$ of the 1D Hubbard model on that subspace. We prove that for Rayleigh–Ritz on any orthogonal projector the relative L_1 error lies between $1 - w$ and $\min\{1 + w, (1 + \sqrt{w})(\sqrt{1 - w} + \hat{\Lambda}(\eta)/\eta)\}$, with w the captured weight and $\hat{\Lambda}$ the boundary coupling at resolution η , assembled from the Ritz pairs without the answer. Weight alone cannot certify: an analytic counterexample forces any bound indexed on it to its trivial value, and on the probe’s support, where $w = 1$ exactly, the error is still 0.35–0.43 at $\eta = 0.18t$ against a ceiling of 2. A subspace containing the order- K Krylov space reproduces the moments through order $2K + 1$, and no further. Measured to dimension 10 306 296, the bound is vacuous at every operating fraction of the resource scan, at all five sizes, by 1.66–8.61, and informative only well above them; it crosses the trivial bound at a true error of $1.2\text{--}7.3 \times 10^{-3}$, though a two-constant rule in $1 - w$ is tighter on 11 of 14 splits. In the determinant basis the probe’s Krylov space has coordinate support on every symmetry-allowed determinant, so at full weight only a subspace holding all of them has zero leakage; for a momentum probe a momentum block, $1/L$ of the sector, does. At full weight a second-order certificate follows; below it, where every operating fraction lies, one is open. One-body fermionic magic does not predict the support in advance: its rank correlation with it is -1 in each of six strata (exact $p = 3.3 \times 10^{-13}$ against independent orderings) but changes sign between families.

I. THE OPERATIONAL QUESTION

A sample-based quantum calculation returns a list of bitstrings. Diagonalizing the Hamiltonian inside the span of the configurations they name yields an energy, and—as this work and several others show—a frequency-resolved response as well. What it does not yield is a statement of how wrong that response is. This paper addresses the question a user is left with once the shots are spent: *given the subspace those shots populated, how wrong is the spectral function built on it?* The answer has a positive part, statements that hold on any subspace, and a negative part, a measurement showing that the resulting certificate is not informative at the sector fractions at which sampling operates.

The positive part consists of three results. First, a two-sided a posteriori bound (Theorem 1): for Rayleigh–Ritz reconstruction on any orthogonal projector, the L_1 error of the broadened spectral function lies between the Born weight of the probe that the subspace failed to capture and that weight plus the Hamiltonian coupling that leaks across the boundary of the subspace at the resolution η , a quantity with a closed form in the Ritz pairs the reconstruction already produces. Second, the captured weight alone cannot play that role: an analytic counterexample forces the constant of any bound indexed on it to its trivial value, and on the probe’s own Born support, where the captured weight is exactly one, the relative error is still 0.43 at $L = 6$ and 0.35 at $L = 8$ at $\eta = 0.18t$ on the periodic rings (0.28 on the open chain of Sec. IV; the two lattices are never pooled). Third, what the subspace structure does guarantee is moment exactness: a subspace containing the order- K Krylov space reproduces the spectral moments through order $2K + 1$ and no further (Theorem 2), and a non-circular count gives the shots that capture every high-probability configuration (Proposition 2).

The negative part is measured. Evaluated on the subspaces of the resource scan, at five sizes up to a sector of dimension 10 306 296, the leakage branch of the bound is worse than the trivial bound at every operating fraction of the resource scan, by factors of 1.66 to 8.61, and at $L = 12$ that verdict is converged in Krylov depth. The reconstructions at those fractions are therefore, at present, uncertified; the one drawn at 85% of its sector, Fig. 3, is certified, loosely (Sec. V). The reconstructions themselves are not in question: an independent implementation recovers the acceptance fractions of the scan to under 2%. Two further measurements locate what remains. The crossing of the trivial bound occurs at a true relative error between 1.2×10^{-3} and 7.3×10^{-3} over sixteen (L, η) cells whose true errors span five decades, so the certificate is a calibrated a posteriori risk tracker, although a two-constant rule in the missed weight,

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fitted out of sample, localizes the same crossing more tightly on eleven of fourteen splits. And in the determinant basis in which the device measures, the Krylov space of the probe has coordinate support on every determinant that the reflection about the seed site allows, measured from $L = 6$ to $L = 14$, so at full captured weight only a subspace holding all of them has zero leakage; a momentum-resolved probe confines that support to one momentum block, $1/L$ of the sector (Sec. VE).

The second half of the paper asks whether the certificate can be replaced by an a priori predictor computed from the state before any circuit runs. The natural candidates are the efficiently measurable monotones of fermionic non-Gaussianity, and on number-conserving states the one-body member collapses to the single invariant $\mathcal{F}_1 = 4 \text{tr}[\gamma(1 - \gamma)]$ (Eq. (16)). An invariant of the occupation spectrum cannot see a boundary: along a geminal orbit every order of it is fixed while the determinant support runs from $O(K)$ to 2^K at bond dimension 2 (Theorem 3), and on fibers where every invariant and the captured weight are frozen the subspace-aware certificate still separates a non-vacuous value from the trivial bound (Proposition 1). Within the six (L, N) strata of the Hubbard interaction sweep the rank correlation of $\mathcal{F}_1$ with the determinant support is exactly -1 (exact permutation $p = 3.3 \times 10^{-13}$ against independent orderings), a deterministic monotone relation whose sign reverses between families, as an orbital invariant compared with a basis-dependent count must (Sec. VI). Sections VII and VIII then price the method in support, resolution and shots, and check the full-sector references against the opposite finite-size scaling of the charge and spin gaps of the Hubbard chain.

Four bodies of work bound what is new here. (i) Patel, Rangi and Tam [1] construct Green’s functions from sampled Krylov subspaces for dynamical mean-field theory [2], independently and concurrently, and cite this work as “a closely related sample-based approach to dynamical spectral functions, constructed directly from bitstring-sampled subspaces”; we make no priority claim against it. (ii) Classical dynamical DMRG arrived first: Benthien and Jeckelmann [3] computed all three momentum-resolved channels reported here, for the half-filled extended Hubbard chain at $L = 90$ – 120 , nineteen years ago, and correction-vector DMRG treats the Hubbard chain at $L = 48$, the Heisenberg chain at $L = 64$ and the t – J ladder at 48×2 sites [4]. (iii) Frequency-resolved responses of correlated lattice models have been measured on devices far larger than ours: a retarded spin Green’s function of the Fermi–Hubbard chain at up to 30 qubits [5], an angle-resolved-photoemission-like $A(k, \omega)$ on a 27-site chain at 54 qubits [6], $S(q, \omega)$ benchmarked against a copper-sulfate crystal [7] and on 50 qubits against neutron scattering in KCuF_3 [8], and quench spectroscopy of $L = 101$ spins [9]. Our device run spans its entire sector, so its agreement with exact diagonalization is exact by coverage rather than a test of the method (Sec. IX). (iv) The division between Gaussian-invariant and basis-dependent quantities is due to Tarabunga *et al.* [10], six weeks before this work first appeared, and to Leone and Bittel [11], seventeen days before; the priority for it is theirs, and what this work adds is a constructive instance and its extension from the determinant support to the leakage. *We claim no quantum advantage in this paper*: the one quantitative statement it makes about its own cost—that the basis-dependent bond dimension tracks the determinant support—argues, wherever that bond dimension is small, for a tensor network rather than for a sampler (Sec. IX).

The paper is organized as follows. Section II fixes the sampling primitive and states which reconstructions carry a shot budget and which are full-sector references. Section III proves the bound and shows that no bound on the captured weight alone can replace it; Sec. IV states what the subspace structure certifies; Sec. V evaluates the certificate; Sec. VI addresses a priori predictors; Secs. VII and VIII price the method and validate the references; and Sec. IX reports the device runs, the molecular suite and the region without advantage. The systems are the half-filled one-dimensional Hubbard model at $U/t = 8$, rings and open chains at $L = 4$ to 14 , and nineteen molecular active spaces. Three different counts share the symbol $\mathcal{S}$ and are kept apart: the static ground-state support of Sec. VI, the dynamical support of Sec. VII, and the set of configurations the sampler returned, the $\mathcal{S}$ of Theorem 1. Proofs, full tables and extended discussions are in the Supplemental Material, cited as Sec. *Sn*.

II. ONE SAMPLING PRIMITIVE, FOUR CHANNELS: WHAT IS RECONSTRUCTED FROM SHOTS, AND WHAT IS NOT

This section states the measurement primitive, fixes the one definition the rest of the paper turns on, and says, channel by channel, which objects here were reconstructed from a finite number of measurements and which are full-sector references computed classically.

A. Construction

For a Hamiltonian $\hat{H}$ with N -electron ground state $|0\rangle$ of energy E_0 , the single-particle spectral function has an electron-addition and an electron-removal branch,

$$A_p(\omega) = \sum_n |\langle n^{N+1} | \hat{c}_p^\dagger | 0 \rangle|^2 \delta(\omega - (E_n^{N+1} - E_0)) + \sum_m |\langle m^{N-1} | \hat{c}_p | 0 \rangle|^2 \delta(\omega - (E_0 - E_m^{N-1})), \quad (1)$$

energies measured from E_0 ; the chemical potential $\mu = \frac{1}{2}(E_0^{N+1} - E_0^{N-1})$ places the Fermi level mid-gap for the particle-hole-symmetric Hubbard model at half filling. The subspace is built in three steps. (1) Prepare the response seeds $|\phi^\pm\rangle = \hat{c}_p^\dagger|0\rangle, \hat{c}_p|0\rangle$ in the $(N\pm 1)$ sectors (every simulation here takes $|0\rangle$, the seeds and $e^{-i\hat{H}t_k}$ exact; how the non-unitary seed is prepared on a device is not specified by this work). (2) Time-evolve each, $|\phi^\pm(t_k)\rangle = e^{-i\hat{H}t_k}|\phi^\pm\rangle$, and measure computational-basis bitstrings at each of the $K+1$ snapshots; the union of the *distinct* bitstrings returned is the determinant subspace $\mathcal{S}^\pm$. (3) Form $\hat{H}_{\mathcal{S}^\pm} = P_{\mathcal{S}^\pm} \hat{H} P_{\mathcal{S}^\pm}$, diagonalize it as $\sum_m \tilde{E}_m^\pm |\tilde{m}^\pm\rangle \langle \tilde{m}^\pm|$, and evaluate the Lehmann sum inside $\mathcal{S}^\pm$. The retarded Green's function is then assembled entirely on the classical processor,

$$G_{pq}(\omega) = \sum_m \frac{\langle 0 | \hat{c}_p | \tilde{m}^+ \rangle \langle \tilde{m}^+ | \hat{c}_q^\dagger | 0 \rangle}{\omega - (\tilde{E}_m^+ - E_0) + i\eta} + \sum_m \frac{\langle 0 | \hat{c}_q^\dagger | \tilde{m}^- \rangle \langle \tilde{m}^- | \hat{c}_p | 0 \rangle}{\omega - (E_0 - \tilde{E}_m^-) + i\eta}, \quad (2)$$

whose imaginary part is the Lorentzian-broadened $A_{pq}(\omega)$ at resolution η . The seed selects the channel: charged seeds $\hat{c}_q^\dagger|0\rangle$ in the $(N\pm 1)$ sectors give $A(\omega)$ and, by Fourier transform of G_{pq} , $A(k, \omega)$; number-conserving density and spin seeds $\hat{n}_q|0\rangle, \hat{S}_q^z|0\rangle$ in the neutral N sector give $S(q, \omega)$ and $S^{zz}(q, \omega)$. The processor measures *only* bitstrings. Figure 1 schematizes the primitive — what one circuit and one readout can address; Table I states separately what was actually run for each channel here.

B. Relation to quantum subspace and Krylov methods

In exact arithmetic the span of the time-evolved seeds is the order- K polynomial Krylov space [15, 16], and diagonalizing $\hat{H}$ inside a subspace containing it is the route of quantum subspace expansion [17], filter diagonalization [18] and real-time quantum Krylov algorithms [19], whose error theory is now characterized [20]. What distinguishes the present method is that these states are used not to *estimate matrix elements* [21–26] — the step requiring Hadamard tests or quantum equation-of-motion measurements on the device — but as a *sampling distribution* selecting a determinant subspace [27–33], extended since to embedded fragments and to auxiliary-field post-processing [34, 35], and here lifted to the $(N\pm 1)$ manifold and to a frequency-resolved observable. That lineage stops at energies and, at most, discrete levels: the closest published results are a QSCI quasiparticle band structure returning discrete poles [36], a molecular spectrum built from a neutral-sector autocorrelation with classically projected long-time dynamics [37], and, from the measured-overlap side rather than the sampled one, a dynamical structure factor of a Kitaev spin liquid obtained by quantum subspace expansion [38], or a hybrid quantum-classical framework for many-fermion response and structure [39]. Closest on lattice models, Nogaki *et al.* [40] apply symmetry-adapted SQD to ground-state *energies*, and Wray *et al.* [41] measure how SQD converges on a variable-length cuprate chain. The readout is that of SQD, the circuits are those of sample-based Krylov diagonalization [29, 42, 43], and all dynamical information is assembled classically through Eq. (2) by one continued-fraction resolvent [12, 44].

C. What the subspace reconstruction is: Rayleigh–Ritz, not restriction

Step (3) is the definition the rest of this paper turns on, and it is stated before it is used. With $P_{\mathcal{S}}$ the orthogonal projector onto the span of the determinants in $\mathcal{S}$ and ϕ the seed,

$$A_{\mathcal{S}}(\omega) = \frac{\eta}{\pi} \left\| (\omega + E_0 + i\eta - H_{\mathcal{S}})^{-1} P_{\mathcal{S}} \phi \right\|^2, \quad H_{\mathcal{S}} = P_{\mathcal{S}} \hat{H} P_{\mathcal{S}}, \quad (3)$$

TABLE I. **Provenance of every dynamical object in this paper: the subspace it was built on, and the measurements that built it.** “Full sector” means no subspace selection of any kind — the continued fraction runs on the whole sector at the stated recursion depth n_ℓ , so the row is a reference against which reconstructions are scored, not a reconstruction. “Observed bitstrings” means the subspace is the union of the distinct bitstrings actually returned by T draws split evenly over the $K+1$ snapshots, which is the protocol of Sec. II; one *stream* is one independent pseudorandom realization of that draw. “Born ranking” means the subspace is the top fraction of the sector under the accumulated Born weight of the evolved state, which is the $T \rightarrow \infty$ limit of the same protocol and carries no shot budget. Sector dimensions are exact; budgets are per channel and per stream.

Channel and system	Shown in	Determinant subspace	Shots T
$A(\omega)$; Hubbard $L=6$, $U/t=8$, $\eta=0.15t$; dim = 300	Fig. 2	observed bitstrings; $ \mathcal{S} = 245$ of 300 in panel (a), 242.5 on average in panel (b)	5.2×10^4 per stream (4×10^3 at each of 13 snapshots, $K=12$), 8 streams
$A(\omega)$, orbital projected; 19 molecules at dissociation; dim = 2–300	Table S12	observed bitstrings; the fraction $ \mathcal{S} /D$ is the last column of that table	2.16×10^5 per molecule (8×10^3 at each of 27 snapshots, $K=26$), one stream
$A(\omega)$; N_2 along dissociation, CAS(6,6)	Fig. S6(a)	full CAS sector	none
$A(\omega)$; hardware, <code>ibm_fez</code> , $L=6$, $U/t=4$, $\eta=0.15t$; dim = 300	Fig. 9(a)	post-selected from device bitstrings read in reversed bit order, so every retained shot is a device error (Sec. IX A); $ \mathcal{S} = 300$ of 300, the entire sector, so the agreement with the exact curve is exact by coverage and is not a fidelity measurement	3.5×10^5 : 5×10^4 on each of 7 circuits, $k=0$ (the $t=0$ reference) to $k=6$
$A(k, \omega)$; $L=12$, $\eta=0.18t$; dim = 731 808	Fig. 4	full sector, $n_\ell = 150$ per momentum	none
$A(k, \omega)$; $L=8$, $\eta=0.18t$; dim = 3920	Fig. 3(a,b)	Born ranking, 3332 of 3920 per channel, over $K=16$ snapshots	none: (a)–(b) are the $T \rightarrow \infty$ limit; at finite shots the same configuration takes 5.2×10^6 per channel (see text)
$A(\omega)$; $L=8$, $\eta=0.15t$; dim = 3920	Fig. 3(c)	observed bitstrings; sector fraction 0.835 ± 0.0025 at the largest budget	up to 2.60×10^6 over 19 snapshots ($K=18$), 8 streams
$S(q, \omega)$; $L=12$, $\eta=0.18t$; dim = 853 776	Fig. 7	full N -particle sector, $n_\ell = 200$	none
$S^{zz}(q, \omega)$; $L=12$, $\eta=0.10t$; dim = 853 776	Fig. 8	full N -particle sector, $n_\ell = 200$	none
Required sector fraction versus L ; $L=4$ –14, $\eta=0.15t$	Fig. 5(b,c)	both selections, plotted together: Born ranking (open squares) and observed bitstrings (filled circles)	measured: 2.4×10^4 , 2.6×10^5 , 3.1×10^6 , 1.9×10^7 at $L = 6, 8, 10, 12$ (Table IV, 4–6 seeds per size)

the spectral function of the *Rayleigh–Ritz operator* $P_S \hat{H} P_S$, whose poles are the Ritz values $\tilde{E}_m$ of Eq. (2) — not eigenvalues of $\hat{H}$ — and not the restriction of A to a subset of its own eigenpairs. The reproduction repository computes exactly this, at the small sizes by forming and diagonalizing the submatrix $\hat{H}[\mathcal{S}, \mathcal{S}]$ and at the large ones by a matrix-free restricted matrix–vector product inside the same continued fraction; the consequence is the mechanism of this paper: *spectral weight the subspace retains is not kept in place, it is moved in ω* . Truncation does not delete Lehmann poles, it displaces every surviving one, so the weight a subspace captures cannot by itself bound how far the reconstruction has moved — which is why no weight-only bound survives contact with Eq. (3) (Sec. III D), and why Theorem 1 carries a second term indexed on the coupling $P_S \hat{H} Q_S$ across the boundary of $\mathcal{S}$, with Q_S the complementary projector—exactly what Eq. (3) discards. Two consequences used later, that determinants outside $\text{supp}(\phi)$ still move the Ritz spectrum and that the error does not vanish on $\text{supp}(\phi)$ itself, are stated in Sec. S1 and measured in Sec. III D.

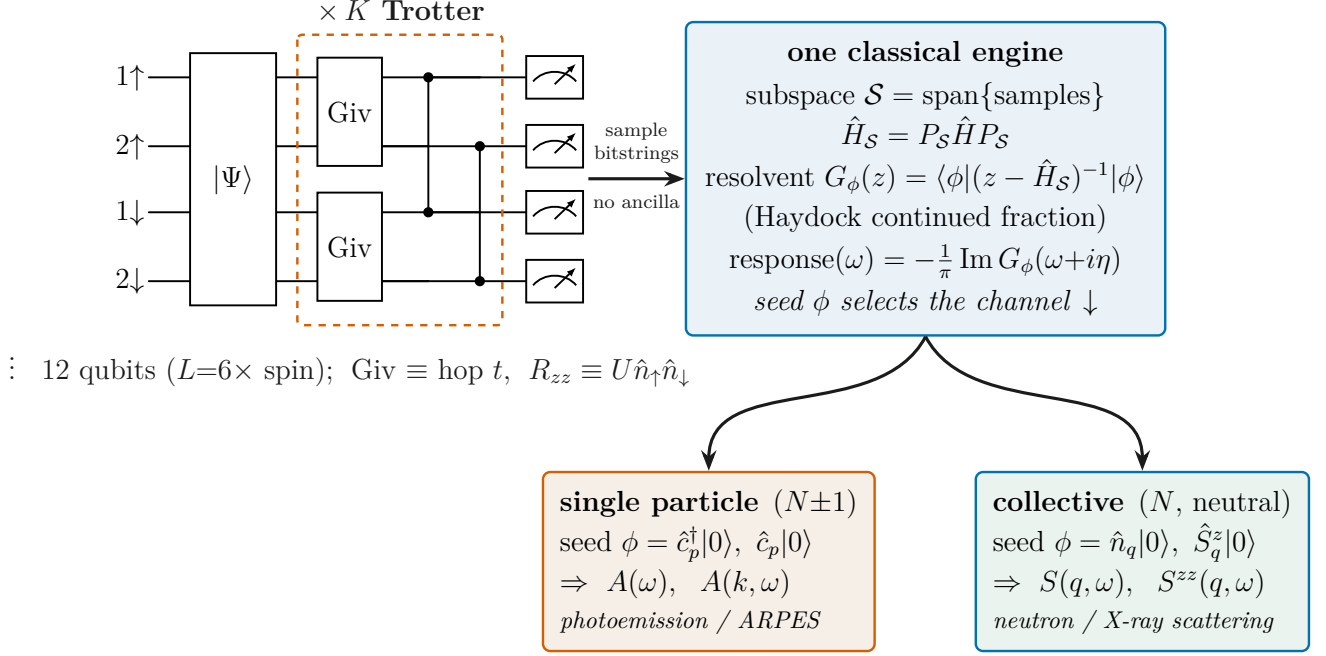


FIG. 1. **One sampling primitive, four dynamical responses.** An ancilla-free circuit—state preparation and a Trotter layer repeated K times (Givens rotations for the hopping t , native R_{zz} for the on-site $U \hat{n}_{\uparrow} \hat{n}_{\downarrow}$)—is measured in the computational basis; no Hadamard test or controlled unitary is run on the device. The sampled bitstrings define a sector-specific subspace $\mathcal{S}$ and feed a single classical engine—a Haydock continued-fraction resolvent [12] $G_{\phi}(z) = \langle \phi | (z - \hat{H}_{\mathcal{S}})^{-1} | \phi \rangle$, whose single-particle case ($\phi = \hat{c}_p^{\dagger} |0\rangle$) is the Green’s function of Eq. (2). The seed ϕ then selects the channel: particle addition/removal seeds $\hat{c}_p^{\dagger} |0\rangle, \hat{c}_p |0\rangle$ in the $(N \pm 1)$ sectors give the single-particle spectral functions $A(\omega)$ and $A(k, \omega)$ (photoemission/ARPES), while number-conserving density and spin seeds $\hat{n}_q |0\rangle, \hat{S}_q^z |0\rangle$ in the N sector give the dynamical structure factors $S(q, \omega)$ and $S^{zz}(q, \omega)$ (neutron [13] and resonant inelastic X-ray [14] scattering)—all from *one* sampling primitive, the seed selecting the sector and hence the subspace. The four channels are realized in Figs. 2 ($A(\omega)$), 4 ($A(k, \omega)$), 7 ($S(q, \omega)$), and 8 ($S^{zz}(q, \omega)$).

D. Provenance: what was reconstructed from shots, channel by channel

Table I gives, for every dynamical object in this paper, the subspace it was built on and the measurements that built it. The neutral channels $S(q, \omega)$ and $S^{zz}(q, \omega)$, the $A(k, \omega)$ of Fig. 4 and the N_2 series of Fig. S6 are full-sector references computed by continued fraction, not reconstructions: the primitive targets those channels, and targeting a channel is not reconstructing it from measurements. The Born ranking of Fig. 3(a,b) and of the published curve of Fig. 5 is the infinite-shot limit of the declared protocol—neither a clairvoyant oracle nor a finite-shot experiment—and its finite-shot replacement is measured with its budget: at $L = 8$, 2.60×10^6 shots over 19 snapshots reach sector fraction 0.835 ± 0.0025 at relative L_1 error $(4.81 \pm 0.27) \times 10^{-3}$ (Fig. 3(c)), and the literal configuration of Fig. 3(a,b) takes 5.2×10^6 shots per channel. The branch sum rules are identities of the ground state, not measurements of a reconstruction; what is guaranteed in advance is moment exactness (Theorem 2) and a shot count for high-probability configurations (Prop. 2), and the second does not imply the first. The full declarations are in Sec. S1.

III. TWO BUDGETS: THE MISSED WEIGHT AND THE BOUNDARY

The object a sample-based calculation delivers is a subspace, and the object a user wants is an error bar on the response built inside it. This section proves the bound that connects them. It is two-sided, it holds for *any* orthogonal projector and therefore for any set of configurations a sampler might return, and its upper branch separates into two terms with different economics: the first is fixed by the captured weight and vanishes once the probe’s support is held; the second is not controlled by the weight, although it falls as the subspace grows. Everything after it is an evaluation of that separation — Sec. V measures where the resulting certificate says something, Sec. VI shows that no invariant of the state can replace it, and Sec. VII prices both terms in shots.

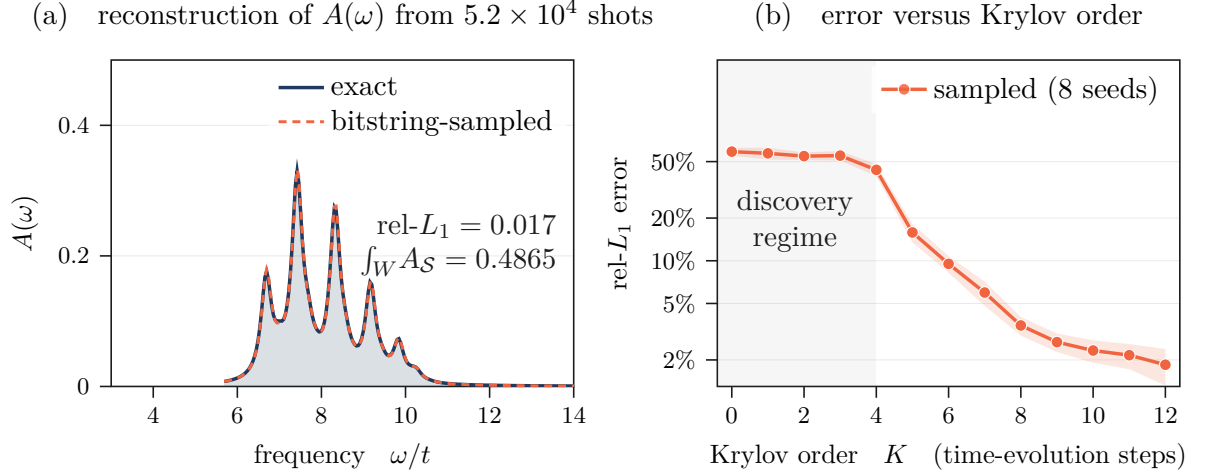


FIG. 2. **The method validated at $L = 6$, and its error against Krylov order.** (a) The bitstring-sampled single-particle spectral function reproduces the exact $A(\omega)$ (Hubbard $L = 6$, $U/t = 8$; broadening $\eta = 0.15t$) to relative- L_1 0.017—the exact (filled) and reconstructed (dashed) curves are indistinguishable. The analytic addition-branch sum rule $\int A^+ d\omega = 1 - \langle n_p \rangle = 0.5$ is an identity of the ground state and not a check on the reconstruction; integrated over the plotted window the exact and reconstructed curves give 0.486588 and 0.486471, agreeing with each other to 2.4×10^{-4} relative and both falling 2.7% short of 0.5 because Lorentzian tails leave the window (Sec. S7). (b) Convergence of the relative- L_1 error with the Krylov order K (number of time-evolution steps sampled), seed-averaged over eight independent runs (band = standard deviation), $\eta = 0.15t$, so the budget is $T = 4000(K+1)$ and reaches 5.2×10^4 at $K = 12$. The error falls by a factor 32 across the thirteen points, from 0.59 at $K = 0$ to 0.0185 at $K = 12$, with one non-monotone step at $K = 3$ inside the discovery phase, while the mean subspace returned grows from 75 to 242 determinants of the 300-dimensional sector. No geometric rate is fitted to the window $4 \leq K \leq 9$ here (Sec. IV B): one is not implied by moment exactness, and inside that window the decay in K is confounded with the growth of $|S|$.

A. The objects

Fix the $(N \pm 1)$ sector, let H be the Hamiltonian restricted to it, E_0 the ground-state energy of the N -electron sector, and let ϕ be the probe — $\hat{c}_p^\dagger |\Psi_0\rangle$ or $\hat{c}_p |\Psi_0\rangle$ for the charged channels, $\hat{n}_q |\Psi_0\rangle$ or $\hat{S}_q^z |\Psi_0\rangle$ for the neutral ones. Write $z = \omega + E_0 + i\eta$ and $R(z) = (z - H)^{-1}$. The η -broadened spectral function of Eq. (2) is

$$A(\omega) = \frac{\eta}{\pi} \|R(z)\phi\|^2, \quad \text{so that} \quad \int_{\mathbb{R}} A(\omega) d\omega = \|\phi\|^2 \quad (4)$$

for every $\eta > 0$, the sum rule holding because the Poisson kernel integrates to one on each Lehmann pole. Let P be an orthogonal projector on the sector, $Q = I - P$, and

$$\phi_P = P\phi, \quad \phi_Q = Q\phi, \quad w_P = \frac{\|\phi_P\|^2}{\|\phi\|^2}, \quad H_P = PHP, \quad R_P(z) = (z - H_P)^{-1}|_{\text{ran } P}, \quad (5)$$

with Ritz pairs $H_P|\tilde{m}\rangle = \tilde{E}_m|\tilde{m}\rangle$ and $c_m = \langle \tilde{m}|\phi \rangle$. The reconstruction of Eq. (3) is $A_P(\omega) = (\eta/\pi) \|R_P(z)\phi_P\|^2$, and it obeys the same sum rule with $\|\phi_P\|^2$ on the right. When $P = P_S$ is diagonal in the determinant basis — the only case a device produces — we write S for the configuration set and w_S, Λ_S for the quantities below.

The magnitude the theorem is indexed on is the Hamiltonian coupling that crosses the boundary of the subspace, averaged over the resolution window rather than over the spectrum. With $g_m = QH|\tilde{m}\rangle$,

$$\Lambda_P(\eta)^2 := \frac{\eta}{\pi} \int_{\mathbb{R}} \|QH R_P(z)\phi_P\|^2 d\omega = \sum_{m,m'} c_m \bar{c}_{m'} \langle g_{m'}, g_m \rangle \frac{2i\eta}{(\tilde{E}_{m'} - \tilde{E}_m) + 2i\eta}, \quad (6)$$

the second equality being exact and proved below; the sum is real, the (m, m') and (m', m) terms being complex conjugates. Two features of Eq. (6) carry the practical content of this paper. It is assembled from objects the

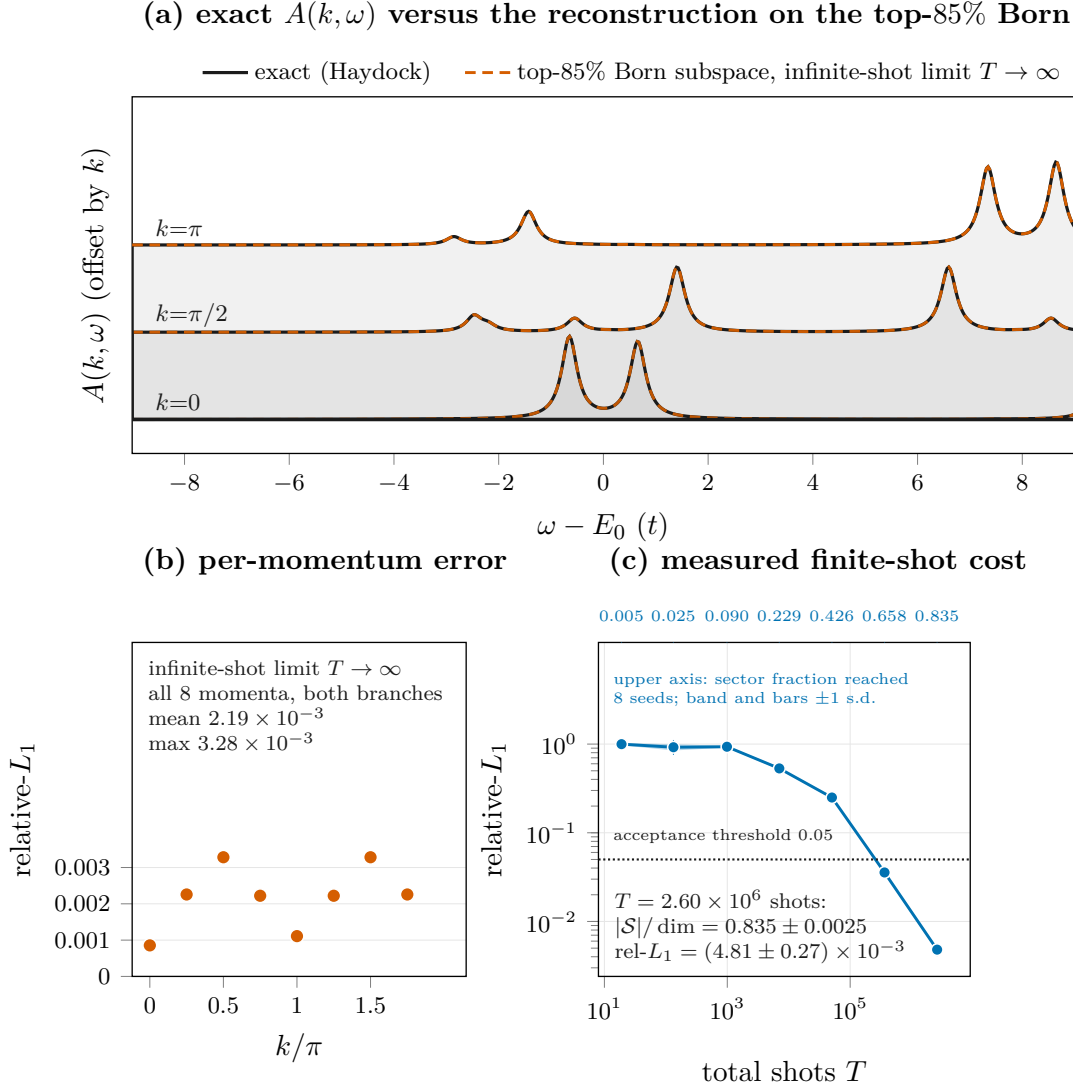


FIG. 3. **The momentum-resolved spectral function on a Born-ranked subspace, and what that subspace costs in shots.** (a) Exact Haydock $A(k, \omega)$ (solid, shaded) and the reconstruction obtained by diagonalising the top-85% of the $(N \pm 1)$ sector under the time-averaged Born weight of the evolved state (dashed), at three momenta ($k = 0, \pi/2, \pi$; Hubbard chain, $L = 8$, $U/t = 8$, $\eta = 0.18t$, $K = 16$ time slices, $\delta t = 0.5$; curves offset vertically by k). The subspace is 3332 of the 3920 determinants of each channel. The ranking in `src/sampled_akw.py` is taken on the *exact* Born distribution, so panels (a) and (b) are the **infinite-shot limit** $T \rightarrow \infty$ of the protocol, not a finite-shot experiment; no shot budget is attached to them. (b) Relative- L_1 error of that reconstruction at all 8 physical momenta, addition and removal branch summed: mean 2.19×10^{-3} , maximum 3.28×10^{-3} . (c) What a finite shot budget buys, measured. Same chain and the same sector dimension ($L = 8$, $\dim(N+1) = 3920$) but a separate run: local seed $c_{0\uparrow}^\dagger |\Psi_0\rangle$, $\eta = 0.15t$, $K = 18$ time slices, T shots split evenly over the 19 snapshots, subspace = the support observed in those T shots, 8 independent seeds. Markers are the seed mean; the shaded band and the error bars are ± 1 s.d. over the 8 seeds (at most 8.1% of the mean at any budget, hence thinner than the line). Upper axis: the sector fraction $|\mathcal{S}|/\dim$ reached at that budget; the dotted line is the acceptance threshold $\text{rel-}L_1 = 0.05$ used by the resource scan. At the largest budget measured, $T = 2.60 \times 10^6$ shots, the observed subspace covers 0.835 ± 0.0025 of the sector and the reconstruction reaches $\text{rel-}L_1 = (4.81 \pm 0.27) \times 10^{-3}$. **The shot budget of (a)–(b), measured.** Panel (c) is at another resolution and seed, so the configuration of (a)–(b) was run at finite shots separately: same $\eta = 0.18t$ and $K = 16$, all 8 momenta, both branches, the subspace being the determinants observed in T multinomial draws per channel, 4 seeds, 32 (channel, seed) pairs. At $T = 5.2 \times 10^6$ shots per channel— 8.3×10^7 over the 16 channels—the reconstruction reaches $\text{rel-}L_1 = (2.11 \pm 0.26) \times 10^{-3}$, maximum 2.57×10^{-3} : panel (b)’s accuracy. It reaches it on a *larger* subspace, covering 0.882 of the addition sector against the 0.850 of (a)–(b), so the budget buys that accuracy together with three extra points of fraction, and not the accuracy at the fraction this figure announces. At half the budget, $T = 2.6 \times 10^6$, the observed subspace covers 0.839—just under the 0.850 of (a)–(b)—and the error is $(6.12 \pm 0.50) \times 10^{-3}$, a factor 2.80 above (b). That factor, and not the 2.05 transferred from Table S6 ($L = 8$, $\eta = 0.15t$, $|\mathcal{S}| = 2180$), is the finite-shot penalty at this configuration; the transferred value understated it by 27%.

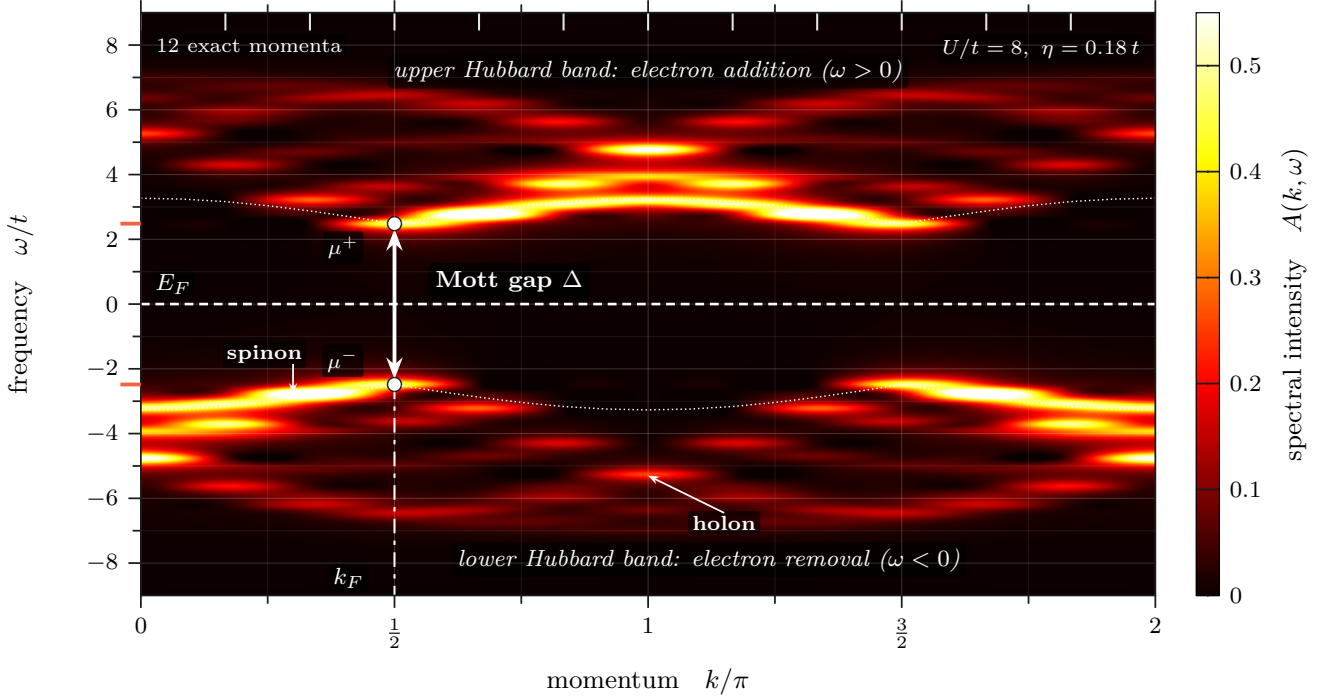


FIG. 4. Momentum-resolved spectral function: the dynamical target. The single-particle spectral function $A(k, \omega)$ of the half-filled Hubbard chain ($U/t = 8$, $L = 12$, Lorentzian broadening $\eta = 0.18t$), computed from the resolvent $\langle \phi | (z - \hat{H})^{-1} | \phi \rangle$ by a Haydock continued fraction [12] of depth $n_\ell = 150$ over the full $(N \pm 1)$ sectors—a finite-depth continued fraction, that is a 150-node Gauss quadrature of the spectral measure [44] and not the Lehmann sum, so the word *exact* is not used of it here (Sec. S7)—the frequency- and momentum-resolved observable that sample-based diagonalization targets but its energy-only output cannot itself deliver. The two dispersing Hubbard bands—electron removal (lower, $\omega < 0$; photoemission) and electron addition (upper, $\omega > 0$; inverse photoemission)—are separated by the correlation-driven *Mott gap* $\Delta = \mu^+ - \mu^- = 4.97t$ between the highest occupied state (μ^- , lower edge) and the lowest unoccupied state (μ^+ , upper edge), obtained from the exact $(N \pm 1)$ ground-state energies (ticks at $\pm 2.484t$); the broadened spectral peaks of the plotted $A(k_F, \omega)$ sit slightly wider, at $\pm 2.49t$ (a $4.99t$ splitting), the $\sim 0.02t$ excess being the $\eta = 0.18t$ Lorentzian broadening; the thermodynamic Lieb–Wu value [45] is $4.68t$, the $L = 12$ excess being finite-size. The gap is *direct* at the Fermi momentum $k_F = \pi/2$. Particle–hole symmetry places the Fermi level at mid-gap ($\omega = 0$), verified to double-precision round-off, and each momentum obeys the analytic spectral sum rule $\int A(k, \omega) d\omega = 1$ (the plotted broadened grid recovers it to $\sim 2\%$ within the $\pm 9t$ window). In the thermodynamic limit the lower Hubbard band is spin–charge separated; at $L = 12$ the plot is compatible with a narrow spinon-like branch (bandwidth $\pi J/2 = 0.79t$ [46], $J = 4t^2/U$; dashed, validated against the exact peak dispersion to $< 0.1t$) rides a broad holon (charge, $\sim 4t$) band, as expected from the Bethe-ansatz solution [45], and is not offered as evidence of it. At $L = 12$ that structure is three to seven poles per momentum (Table S7); the words appropriate to the thermodynamic limit are not used for it. The momentum axis is a periodic cubic-spline interpolation of the 12 physical momenta (ticks, top). Faithfully *sampling* this full momentum-resolved observable—as opposed to the ground-state energy—requires a large fraction of the $(N+1)$ sector (Fig. 5), the cost quantified in Sec. VII.

reconstruction has already produced — the Ritz vectors and their coefficients — plus one matrix–vector product $H|\tilde{m}\rangle$ per Ritz vector, so evaluating it costs a small multiple of the diagonalization that produced A_P and *does not require* A . And it vanishes exactly when $\text{span}(S)$ contains $\mathcal{K}(H, \phi_P)$, the Krylov space of the retained part of the probe: $QHR_P(z)\phi_P$ vanishes for every z exactly when $\mathcal{K}(H_P, \phi_P)$ is invariant under H , and it then coincides with $\mathcal{K}(H, \phi_P)$. For a subspace that holds the whole probe, $w_P = 1$, the condition is that $\text{span}(S)$ contain $\mathcal{K}(H, \phi)$ itself—far more demanding than it looks, as Sec. V E measures.

B. The bound

Theorem 1 (a posteriori leakage certificate). *Let $H = H^\dagger$ act on the $(N \pm 1)$ sector, let $\eta > 0$, let P be any orthogonal projector on that sector, and let A , A_P , w_P and $\Lambda_P(\eta)$ be as in Eqs. (4)–(6). Then*

$$\|\phi\|^2(1 - w_P) \leq \|A - A_P\|_1 \leq \min \left\{ \|\phi\|^2(1 + w_P), (\|\phi\| + \|\phi_P\|) \left(\underbrace{\|\phi_Q\|}_{\text{missed weight}} + \underbrace{\Lambda_P(\eta)/\eta}_{\text{boundary leakage}} \right) \right\}, \quad (7)$$

equivalently, dividing by $\|\phi\|^2$ and writing $\hat{\Lambda}_P = \Lambda_P/\|\phi\|$,

$$1 - w_P \leq \frac{\|A - A_P\|_1}{\|\phi\|^2} \leq \min \left\{ 1 + w_P, (1 + \sqrt{w_P})(\sqrt{1 - w_P} + \hat{\Lambda}_P(\eta)/\eta) \right\}.$$

Three hypotheses carry it, each a property of the *pipeline* that produced A_P rather than of the inequality: (H1) A and A_P are referred to the same E_0 ; (H2) the L_1 norm is taken over the whole line and normalized by $\|\phi\|^2$; (H3) A is the exact spectral function of the full sector. Each is load-bearing and each is checked against this paper’s own pipeline in Sec. S11; the proof, from the Feshbach–Löwdin split of the resolvent (Eq. (S1)) and Cauchy–Schwarz in ω , is in Sec. S2.

C. What the two branches are, and what they are not

What the weight does not control. The first summand of the leakage branch, $\|\phi_Q\| = \|\phi\|\sqrt{1 - w_P}$, is the Born weight of the probe the subspace failed to hold: it vanishes once the sampler holds the probe’s Born support, and Table I prices that purchase channel by channel. The second, $\Lambda_P(\eta)/\eta$, reads the off-diagonal block PHQ , and the captured weight does not control it: along the sampler’s ranking it falls as S grows (Sec. V), but a subspace can hold the whole probe and still leak. Section IIID turns this from a reading into a measurement: on $\mathcal{S}^* = \text{supp}(\phi)$, where the first summand vanishes identically and by construction, the relative L_1 error remains of order one, against a ceiling of 2.

A certificate, not an error estimate. The gap between the bound and the measured error is large, and it is not the accident of a constant. At *fixed* P the looseness is stable — within $\pm 17\%$ ($L = 6$) and $\pm 25\%$ ($L = 8$) over a decade in η — but across subspaces it varies by a factor 51, and at the threshold where the leakage branch first improves on the trivial one it is 3×10^2 to 2.4×10^3 (Sec. VD). Eq. (7) is therefore an exclusion, not an error estimate; Sec. V reports what it excludes, and at what fraction of the sector.

Only one of the two terms is a posteriori. $\Lambda_P(\eta)$ is computable from H , $\text{ran } P$, ϕ and the Ritz vectors alone. The weight term is not: w_P needs the exact decomposition of the probe norm, and estimating it against an already-truncated Ψ_0 is biased in the optimistic direction. The complete bound is therefore *not* evaluable by a user who does not already have the answer, and we never claim that it is; Sec. VD gives the conservative substitution that makes the decision itself a posteriori in the regime where it is used.

Prior art, and what is new. Nothing in the proof is new as analysis. The split is Löwdin partitioning [47], the algebra of the Feshbach projection [48]; Rayleigh–Ritz and Krylov error analysis for resolvents and matrix functions [20, 44, 49–54] contains every rung of its steps, and the integration step is the reduced-basis bound on a Galerkin projection with inf–sup constant η [55, 56]. A posteriori estimates for Krylov approximations to matrix functions are due to Frommer and Simoncini, Jia and Lv, and Frommer and Schweitzer [57–59], with Lanczos and stochastic-Lanczos-quadrature versions [60–62]; on quantum Krylov, Oliveira, Ziemas and Glaser assess reliability without knowledge of the true spectrum [63]; and a response-targeted selection criterion for classical selected configuration interaction is Reinholdt, Kjellgren and Kongsted’s [64]. The priority for the a posteriori framing is theirs. What we add is the conjunction of three things: a closed form for $\Lambda_P(\eta)$ from the Ritz pairs on a subspace of *coordinates*, with no Krylov structure, invariance or gap assumption; a bound two-sided on the same object; and a measured calibration of where it crosses the trivial bound (Sec. VD). Whether it is useful at the fractions sampling reaches is answered, in the negative, in Sec. V; the full comparison with that literature is in Sec. S2.

D. The captured weight is not the controlling magnitude

The certificate one would want for a sampled reconstruction indexes the error on the captured weight alone,

$$\|A - A_S\|_1 \leq 2\|\phi\|^2(1 - w_S) + C\rho^{-K_S}, \quad (8)$$

with $w_S = \|P_S\phi\|^2/\|\phi\|^2$ and K_S the largest Krylov order contained in $\text{span}(S)$ (we set $K_S = -1$ when $\phi \notin \text{span}(S)$). It is false, for every constant C and every ρ , and it fails hardest in exactly the regime a sampling experiment aims for.

A two-level counterexample forces the trivial constant. Take $H = \begin{pmatrix} 0 & g \\ g & 0 \end{pmatrix}$, $\phi = e_0$, $S = \{0\}$, $E_0 = 0$. Then $P_S\phi = \phi$, so $w_S = 1$ exactly and the weight term vanishes identically; and $\mathcal{K}_0 = \text{span}\{\phi\} \subseteq \text{span}(S)$ while $\mathcal{K}_1 \not\subseteq \text{span}(S)$, so $K_S = 0$ exactly and the second term is the bare constant C . The exact A is a pair of Lorentzians of weight $\frac{1}{2}$ at $\pm g$; the reconstruction, built on $H_S = P_S H P_S = 0$, is one Lorentzian of weight 1 at the origin, so $\|A - A_S\|_1 \rightarrow 2\|\phi\|^2$ as $g/\eta \rightarrow \infty$. Since g is free, (8) requires $C \geq 2\|\phi\|^2$: false if C is a constant, vacuous if it may depend on the instance. The same holds at every Krylov order: on a chain of $K + 2$ sites with $\phi = e_0$ and S its first $K + 1$ sites, $w_S = 1$ and $K_S = K$, and the exact and reconstructed spectra have strictly interlacing poles, so the same limit requires $C\rho^{-K} \geq 2\|\phi\|^2$ for every K , which no fixed C satisfies when $\rho > 1$. With $w_S < 1$ strictly the weight term alone fails without bound: in a three-level family its violation diverges as ε^{-2} in the amplitude ε of the discarded component (Sec. S9). By contrast $\|A - A_S\|_1 \leq \|\phi\|^2(1 + w_S)$ holds for every S under no hypothesis at all, A and A_S being nonnegative with known integrals: that trivial bound is the only surviving member of the family (8), and Theorem 1 keeps it explicitly, as the first branch of its minimum.

The violation is systematic, and worst where the reconstruction is best. In a pooled sweep of 606 subspaces from five Hamiltonian families the weight-only bound fails on 468, and on all 363 with $w_S \geq 0.99$, the mildest by a factor 2.10 (Fig. S5); in a first sweep it failed on 126 of 126 such subspaces. In the configuration of Fig. 3 its median violation factor rises from 18.5 to 29.9 between sector fractions 0.50 and 0.95 while the measured error falls by three orders of magnitude. The sweep and its declared limits are in Sec. S2; the conclusion does not depend on it, the counterexample being analytic.

Why it fails. The reconstruction is Rayleigh–Ritz on $H_S = P_S H P_S$, not a restriction of A to a subset of its eigenpairs: weight the subspace keeps is *displaced* in ω , not discarded, and the two-level example is the extreme case. What the integrals do give is the lower bound $\|\phi\|^2(1 - w_S) \leq \|A - A_S\|_1$, with 0 violations across the 606 subspaces, the left-hand half of Theorem 1.

The two terms are mutually exclusive, and one is off at every operating point here. S is a set of determinants, so P_S is diagonal in the working basis and $P_S\phi = \phi \iff \text{supp}(\phi) \subseteq S \iff w_S = 1$, while $K_S \geq 0$ means precisely $\phi = H^0\phi \in \text{span}(S)$. Hence the exclusion lemma

$$K_S \geq 0 \iff w_S = 1, \quad (9)$$

verified with 0 discrepancies across the 378 deposited subspaces, of which 357 have $K_S = -1$ (Sec. S9). The two terms of (8) are therefore never both informative: either $w_S = 1$ and the whole bound is the second term — the two-level counterexample — or $K_S = -1$ and $C\rho^{-K_S} = C\rho$, an undefined constant with no decay in it. Counting settles which branch applies here. The seed is $\phi = \hat{c}_{0\uparrow}^\dagger|\Psi_0\rangle$ at half filling, so $\text{supp}(\phi)$ lies in the set of $(N+1)$ determinants with orbital $0\uparrow$ occupied, a fraction $n_\uparrow/L = \frac{1}{2} + \frac{1}{L}$ of the sector, and fills it except where symmetry zeroes an amplitude of Ψ_0 . By (9), $K_S \geq 0$ demands $|S| \geq |\text{supp}(\phi)|$, a floor of 0.571 at $L = 8$ (2237 of 3920), 0.600 at $L = 10$, and above 0.54 at $L = 12$ and 14 at any amplitude threshold from 10^{-12} to 10^{-10} , against the sector fractions 0.5561, 0.3625, 0.1700, 0.0800 at which the resource scan of Sec. VII operates (Fig. 5; the coarser replication grid of Table S5 reads 0.180 at $L = 12$ for the same operating point). From $L = 8$ on, every operating point of that scan lies below its own floor, so $K_S = -1$ by counting alone, with no numerics, no ranking and no protocol, and any point below the same line inherits the verdict (Sec. S9). Theorem 1 carries no such bookkeeping and is held to the opposite standard. Its two terms are defined and simultaneously active for every S ; the weight term is the half this work cannot compute without Ψ_0 — $\|\phi_Q\| = \|\phi\|\sqrt{1 - w_S}$, and estimating w_S against an already-truncated Ψ_0 is biased optimistically — while $\Lambda_S(\eta)$ is an *a posteriori* quantity assembled from the Ritz vectors the reconstruction already forms plus one matrix–vector product, and is evaluated, subspace by subspace, in Sec. V.

The magnitude, not the constant. Two measurements at the same, exactly unit, captured weight show that (8) was indexed not on a bad constant but on the wrong quantity. First, let $S^* = \text{supp}(\phi)$, the smallest determinant subspace of unit captured weight — fixed by the seed, with no ranking, no tie-breaking and no zero-amplitude padding — of size $(\frac{1}{2} + \frac{1}{L})D$ at $L = 6$ and $0.571D$ at $L = 8$, where 426 of the 4900 ground-state amplitudes vanish by symmetry. Its weight term is identically zero, and its relative L_1 error is 0.43 at $L = 6$ and 0.35 at $L = 8$ on the periodic rings at $\eta = 0.18t$, rising to 1.15 and 0.73 at $\eta = 0.05t$, against a hard ceiling of $1 + w_S = 2$. Second, S^* is the $K = 0$ rung of the Krylov ladder $S_K = \bigcup_{a \leq K} \text{supp}(H^a\phi)$, every rung of which contains ϕ and so carries $w_S = 1$ identically, while its moment reach grows: S_K reproduces the exact spectral moments through order $2K+1$ — to the 10^{-15} level, failing as it must at $2K+2$ — whereas coordinate subspaces of the same size drawn at random miss already μ_0 , in all of 300 draws (Sec. IV). Along a family of constant, exactly unit captured weight the error therefore runs from order unity to moment-exact through order $2K+1$. The captured weight, identical on every rung, does not order them.

What does is elementary. Writing μ_j, μ_j^S for the moments of the unbroadened measures of Sec. IV, every determinant subspace with $w_S = 1$ reproduces μ_0 and μ_1 exactly, and the first moment that can fail is μ_2 , with deficit

$$\mu_2 - \mu_2^S = \|Q_S H P_S \phi\|^2, \quad Q_S = I - P_S, \quad (10)$$

the squared coupling of the Hamiltonian across the boundary of S (E_0 drops out because $Q_S\phi = 0$), zero exactly when S also holds $\text{supp}(H\phi)$ — the $K_S \geq 1$ condition. That boundary coupling is the magnitude the replacement is indexed on: it controls the leakage term through $\Lambda_S(\eta) \leq \|P_S H Q_S\| \|\phi_S\|$, and Theorem 1 averages it over the resolution window rather than over the spectrum. The captured weight says what the subspace *contains*; the error is set by what the Hamiltonian does to the seed at the subspace’s *edge*. Both halves of this paper follow from that displacement: Sec. V measures where the boundary term is small enough for the certificate to say something, and Sec. VI shows why no orbital-rotation invariant can predict it in advance.

IV. WHAT THE STRUCTURE DOES CERTIFY: MOMENT EXACTNESS

Section IIID rules out an error bound indexed on the captured weight. It rules out nothing about the construction itself. One structural guarantee does hold, and it is the one that says what a Krylov-adequate subspace actually buys: a finite list of spectral moments, exactly. We state it with the proof the code realizes, measure its exact reach, and separate it from two claims that attach themselves to it and do not follow from it.

A. The statement that survives, and the proof it actually has

Write $\mathcal{K}_K = \text{span}\{\phi, H\phi, \dots, H^K\phi\}$ for the order- K Krylov space of the seed $|\phi\rangle = \hat{c}_p^\dagger|0\rangle$, and let $d\mu$ and $d\mu_S$ be the unbroadened Stieltjes measures whose Lorentzian convolutions are the exact A and the reconstruction A_S , the Lehmann sum evaluated inside $\text{span}(S)$ as in Sec. II (Sec. S10).

Theorem 2 (Moment exactness). *Let P be the orthogonal projector onto $\text{span}(S)$ and $H_S = PHP$. If $\text{span}(S) \supseteq \mathcal{K}_K$, then*

$$\int \omega^j d\mu_S(\omega) = \int \omega^j d\mu(\omega) = \langle \phi | (H - E_0)^j | \phi \rangle, \quad 0 \leq j \leq 2K + 1. \quad (11)$$

Proof. Because $\mathcal{K}_K \subseteq \text{span}(S)$ we have $PH^a\phi = H^a\phi$ for $0 \leq a \leq K$, whence by induction $H_S^b\phi = PH^b\phi$ for $0 \leq b \leq K + 1$. Then for any $a \leq K$ and $b \leq K + 1$, using only $P = P^\dagger = P^2$ and $H = H^\dagger$,

$$\langle \phi | H_S^{a+b} | \phi \rangle = \langle H_S^a \phi | H_S^b \phi \rangle = \langle H^a \phi | PH^b \phi \rangle = \langle PH^a \phi | H^b \phi \rangle = \langle \phi | H^{a+b} | \phi \rangle, \quad (12)$$

and every $j \leq 2K + 1$ is such an $a + b$. Both measures are shifted by the same E_0 , so the identity survives term by term in the binomial expansion of $(H - E_0)^j$. $\square$

The proof is three lines of Rayleigh–Ritz. It does not identify A_S with a Gaussian-quadrature rule, which is not the object built here: every reconstruction diagonalizes $P_S H P_S$ on a coordinate subspace, so $d\mu_S$ carries $|\mathcal{S}|$ nodes, not $K + 1$ (Sec. S3).

Table II measures Eq. (11) on the construction this work uses—the union of the determinant supports of $\{\phi, H\phi, \dots, H^K\phi\}$ —for the open Hubbard chain at $L = 6$, $U/t = 8$, in the $(N+1, S_z=+1)$ sector of dimension $D = 300$. Through order $2K + 1$ the moments agree to a relative 10^{-14} ; at order $2K+2$ they part company by eleven orders of magnitude at $K = 0$ and seven at $K = 1$. The order $2K + 1$ is the exact reach of the construction, not merely a sufficient claim, and the two rows that exhibit that reach are the two on which the ladder has not yet filled the sector.

A negative control fixes what the table is about. Over 300 coordinate subspaces of the *same* size $|\mathcal{S}| = 200$ —two thirds of the sector—drawn uniformly from it, every single draw fails already at the zeroth moment, against 4×10^{-16} for the Krylov-support subspace of identical size (Sec. S10.2). What Theorem 2 certifies is *which* determinants the subspace holds, not *how many*—a structural property, decidable from the subspace alone, before any exact reference exists.

One caveat of scale belongs with the table. The ladder terminates once the subspace contains the cyclic subspace generated by ϕ , and there the reconstruction is exact: at $L = 6$ that is $K = 2$ on the open chain, and already $K = 1$ on the periodic rings of Sec. VII, where it stalls at 296 of 300 determinants with $A_S = A$ to 10^{-14} (Sec. S10.2). Three rows is all this validation system has, and we claim nothing about the large- K behaviour of $\|A - A_{\mathcal{K}_K}\|_1$. Within the reach it covers, the identity is exact and sharp at the next order.

TABLE II. **Moment exactness and its exact reach.** Open Hubbard chain, $L = 6$, $U/t = 8$; seed $|\phi\rangle = \hat{c}_{0\uparrow}^\dagger|0\rangle$; $(N+1, S_z=+1)$ sector of dimension $D = 300$; $\mathcal{S} = \bigcup_{a \leq K} \text{supp}(H^a \phi)$. The largest Krylov order inside $\text{span}(\mathcal{S})$, $K_{\mathcal{S}}$, and the captured Born weight $w_{\mathcal{S}}$ are measured, not assumed. ϵ_j is the *relative* discrepancy of the j -th moment, quoted as the largest value over six equivalent implementations of the same identity; the absolute discrepancy at order $2K+2$ follows in parentheses. Only relative residues are quoted below that order, because μ_j spans five decades between $j = 0$ and $j = 6$ and the absolute roundoff residue is route-dependent (Sec. S10.2). At $K = 2$ the ladder has filled the sector, so $\text{span}(\mathcal{S})$ carries every eigenvector of nonzero Lehmann weight, $A_{\mathcal{S}} = A$ identically, and there is no order- $2K+2$ failure to exhibit; that row is kept to mark where the ladder terminates, and its single entry is the roundoff level of the moments.

K	$ \mathcal{S} $	$ \mathcal{S} /D$	$w_{\mathcal{S}}$	$K_{\mathcal{S}}$	$\max_{j \leq 2K+1} \epsilon_j \longrightarrow \epsilon_{2K+2}$ (absolute)
0	200	0.667	1	0	$1.8 \times 10^{-15} \longrightarrow 9.5 \times 10^{-4}$ (3.10×10^{-2})
1	280	0.933	1	1	$3.2 \times 10^{-15} \longrightarrow 6.0 \times 10^{-8}$ (1.34×10^{-4})
2	300	1.000	1	full sector	$7.5 \times 10^{-15} \longrightarrow$ no failure: $A_{\mathcal{S}} = A$

B. What moment exactness does not buy

Moment exactness is a list of $2K + 2$ linear identities, and the L_1 distance between two Lorentzian-broadened measures is not a finite linear functional of them, so geometric convergence in the Krylov order does not follow from it. We state that claim as what it is.

Conjecture 1 (Geometric convergence in the Krylov order). *Fix $\eta > 0$ and let $A_{\mathcal{K}_K}$ be the Rayleigh–Ritz reconstruction on $\mathcal{K}_K$. Are there $C(\eta) < \infty$ and $\rho(\eta) > 1$, uniform over a class of Hamiltonians containing the models studied here, such that $\|A - A_{\mathcal{K}_K}\|_1 \leq C(\eta) \rho(\eta)^{-K}$ for every K ?*

The two-level family of Sec. IIID already forces $C(\eta) \geq 2\|\phi\|^2$ at $K = 0$, so any content the conjecture has must be carried by $\rho(\eta)^{-K}$ alone. Nor is Eq. (11) an error bound: for a coordinate subspace, containing $\mathcal{K}_K$ forces $w_{\mathcal{S}} = 1$, and at unit captured weight the error is $O(1)$ —0.28 on the open chain of Table II at $\eta = 0.18t$, and up to 0.43 on the periodic rings (Sec. III). The panel of Fig. 2 read against the conjecture, and the full discussion, are in Sec. S3.

C. The shot count

Proposition 2 counts the shots that populate such a subspace. Bitstrings are drawn from the time-averaged Born mixture $p(x) = \frac{1}{K+1} \sum_{k=0}^K |\langle x | e^{-iHt_k} \phi \rangle|^2 / \|\phi\|^2$, and the target is to capture every configuration with $p(x) \geq \varepsilon$ at confidence $1 - \delta$. The natural answer, $T = O(\varepsilon^{-1} \log(|\mathcal{S}|/\delta))$, is circular: $|\mathcal{S}|$ is the random output of the very T draws the bound is meant to fix.

A non-circular form costs nothing. With $\mathcal{N}_\varepsilon = \{x : p(x) \geq \varepsilon\}$, each member contributing at least ε to a probability distribution, $|\mathcal{N}_\varepsilon| \leq 1/\varepsilon$ *deterministically*; a union bound over $\mathcal{N}_\varepsilon$ of the event “ x never drawn in T trials” then gives $\Pr[\mathcal{N}_\varepsilon \not\subseteq \mathcal{S}] \leq \varepsilon^{-1} e^{-\varepsilon T}$, so

$$T \geq \varepsilon^{-1} \log(1/(\varepsilon\delta)) \quad (13)$$

suffices (Prop. 2). This is non-circular, and wherever the sampler returns more than ε^{-1} distinct configurations its logarithm is the smaller of the two as well.

The parameter ε thresholds a Born probability, not a spectral accuracy: even at $w_{\mathcal{S}} = 1$ the error is $O(1)$, so capturing the high-probability configurations is a statement about the sampler, not about the spectrum; what the resulting subspace costs is priced in Sec. VII (Sec. S3 gives the two compositions of the bound this work does not make).

V. WHAT THE CERTIFICATE CAN AND CANNOT ASSERT: THE NON-VACUITY THRESHOLD OF THE LEAKAGE BOUND

Throughout this section FR^* is the *non-vacuity threshold of the certificate of Theorem 1*: the sector fraction above which the leakage branch of the bound first improves on the trivial bound $\|\phi\|^2(1 + w_{\mathcal{S}})$. It is a property of the inequality chain that proves the theorem, not a frontier of the method: at $L = 8$, $\eta = 0.18t$ the acceptance criterion of this paper, a relative L_1 error of 0.05, is reached at $\text{FR} \approx 0.52$, while the certificate does not reach 0.05 until $\text{FR} \approx 0.98$.

Non-vacuity is a condition on one number. With $\hat{\Lambda}_S = \Lambda_S/\|\phi\|$, the leakage branch of Eq. (7) improves on the trivial branch exactly when

$$\frac{\hat{\Lambda}_S(\eta)}{\eta} < \tau(w_S) := \frac{1 + w_S}{1 + \sqrt{w_S}} - \sqrt{1 - w_S}, \quad \tau(1) = 1, \quad (14)$$

so a subspace holding the entire Born weight of the probe certifies something as soon as its leakage, in units of the resolution it was asked for, drops below unity. This section is that one comparison, made 204 times.

A. What is evaluated

We evaluate Theorem 1 on the subspaces of Fig. 5 at $L = 6, 8, 10$ and 12 , with the ranking protocol of Sec. VII, over fractions 0.20 – 0.95 and $\eta = 0.10$ – $0.25t$: 204 evaluations, of which 187 are converged in Krylov depth, and no row flagged not converged is quoted. To that grid we add twenty evaluations at the published subspaces themselves, at the integer counts 246, 2180, 19183, 124407 and 824504 at $L = 6$ – 14 , all passing the depth test. The subspaces are the $T \rightarrow \infty$ limit of the protocol, not finite-shot draws. What is certified is the object the code forms, the reorthogonalized Krylov reconstruction A_K at a declared depth, against a reference whose drift is declared, and no measured error within a factor 20 of that drift is quoted. The protocol, the controls and the costs are in Secs. S4 and S11.

B. The threshold, measured: at every published fraction the certificate is vacuous

Evaluated on the published subspaces themselves, the leakage branch of the bound exceeds the trivial bound $\|\phi\|^2(1 + w_S)$ at all five sizes and all four resolutions, by factors running from 1.66 ($L = 6$, $\eta = 0.25t$) to 8.61 ($L = 14$, $\eta = 0.10t$), and by 2.40 to 4.61 at the $\eta = 0.18t$ of the momentum-resolved channel; Table III carries the twenty values. They are monotone in both arguments—rising with L at every resolution, falling with η at every size—so the trend runs against us: the certificate is furthest from saying anything at the largest sector and the finest resolution, which is where the method is aimed.

The largest classically converged case carries the evidence. At $L = 12$ and the published subspace— $|S| = 124407$ of 731808 determinants— $w_S = 0.998622$, so Eq. (14) asks for $\hat{\Lambda}_S/\eta < \tau(w_S) = 0.9625$; the measured value is 4.236 at $\eta = 0.18t$, converged in depth (Sec. S4). Over the 606 live subspaces of the pooled sweep the leakage branch is the smaller of the two on only 61. **The reconstructions at the operating fractions of the resource scan—Fig. 5 and the operating points of Secs. VII and VIII—are therefore, at present, uncertified.** Fig. 3, drawn at 85% of its sector, is the exception: there the certificate of the depth-500 Krylov reconstruction is non-vacuous in all sixteen channels, at a relative bound of 0.77–1.75 against the trivial 2, and loose, 350–900 times the true error (`cert_akw.json`). The $L = 14$ row rests on the stored ground-state checkpoint and was not re-ranked; its caveats are in Sec. S4.

Two controls come with the table. Applying this paper’s own acceptance criterion to the same sweep, with code sharing neither reference, normalization nor recursion depth with the scan, returns the reported fractions to under 2% (Table S1), and the last column of Table III evaluates the defining criterion at the published subspaces themselves, 0.0463–0.0485 against the threshold 0.05. What fails is the certification of those fractions, not the reconstruction and not the curve. In shots the gap to the non-vacuous band is a budget: reaching it multiplies the shots per snapshot needed for the marginal determinant by 7.5–10 at $L = 8$ and by 20–35 at $L = 12$ (Table S2).

C. The strongest certified statements, and what this paper’s own criterion would cost

What the sweep does certify, at each size the strongest statement whose bracket is converged, is collected in Table S3: at $L = 12$, 60% of a 731808-dimensional sector gives relative $L_1 \leq 0.717$, a factor 2.8 below the trivial cap, tightening to ≤ 0.218 at 70% and $\eta = 0.25t$. At the acceptance criterion this paper uses everywhere else, relative $L_1 < 0.05$, the certificate requires at least 0.98 of the sector at $L = 8$, the crossing lying in $[0.977, 0.983]$, and on our grid it is not reached at all at $L = 6, 10$ or 12 : at the accuracy this paper accepts elsewhere, Theorem 1 does not certify the operating points of Fig. 5, and the shot cost of the regime where it would exceeds the saving that figure reports.

TABLE III. **The certificate on the subspaces this paper publishes, including the size the sweep never reached.** Each row is Theorem 1 evaluated at the integer determinant count $|\mathcal{S}| = \text{round}(\text{frac} \times \text{dim})$ that Fig. 5 reports, not at the nearest point of the grid of Table S1. Columns 5–8 are the vacuity factor—the leakage branch of Eq. (7) over the trivial branch $\|\phi\|^2(1 + w_S)$ —so a value above 1 means the certificate says nothing the trivial bound does not. All twenty evaluations are the $T \rightarrow \infty$ limit of the ranking protocol and pass the depth test; controls and costs are in Sec. S11. The last column is the measured relative L_1 error at the resolution and against the threshold that *define* the published fraction ($\eta = 0.15t$, $\text{rel-}L_1 < 0.05$), from code sharing neither reference, normalization nor recursion depth with the scan that produced the fractions. At $L = 14$ the count is one determinant above the deposited 824 503 and the ranking was not re-run; see the text. The $L = 14$ row is taken at the deepest recursion run, depth 250, where the depth indicator is 0.045 and the factor at $\eta = 0.10t$ still falls by 0.17% between depths 200 and 250 ($8.627 \rightarrow 8.613$); the vacuity is not at risk.

L	$\text{dim}(N+1)$	published subspace		vacuity factor at $\eta/t =$				rel- L_1 at $\eta = 0.15t$
		fraction	$ \mathcal{S} $	0.10	0.15	0.18	0.25	
6	300	0.8200	246	4.54	2.94	2.40	1.66	0.0477
8	3 920	0.5561	2 180	5.65	3.69	3.05	2.17	0.0478
10	52 920	0.3625	19 183	6.50	4.28	3.54	2.52	0.0484
12	731 808	0.1700	124 407	8.04	5.20	4.27	3.00	0.0485
14	10 306 296	0.0800	824 504	8.61	5.59	4.61	3.26	0.0463

D. Where the certificate is informative, and where it is not

One positive result survives the sweep. Across all sixteen (L, η) cells the crossing of the trivial bound occurs at a true relative L_1 error of 1.2×10^{-3} to 7.3×10^{-3} , median 2.8×10^{-3} : a spread of 5.9 over true errors spanning five decades, and in every cell at least 47 times the drift of our own reference. The spread is a drift, not scatter, and it runs the conservative way in L : the larger the system, the smaller the true error at which the certificate first says anything. $\hat{\Lambda}_S(\eta)/\eta$ is therefore a calibrated a posteriori risk tracker, computed from H , S , ϕ and η alone, with four limits (Sec. S4). It locates one error level per cell and is not an error estimate. The comparison needs $\tau(w_S)$, which a user can discharge only conservatively: an assumed floor $w_S \geq 0.995$ gives $\tau \geq 0.9280$, at the cost of at most 7% in the threshold. Sixteen cells are the whole of the evidence, and no extrapolation to $L = 14$ is made. And a two-constant rule in $1 - w_S$, fitted on some sizes and tested on the others, localizes the crossing error more tightly than the certificate on eleven of the fourteen train/test splits (`null_ws7.py`), so we claim no superiority for the certificate as a tracker against the captured weight. What distinguishes it is that it is an inequality with a proof, whose decision survives replacing w_S by a floor, while the fitted rule asks for $1 - w_S$ to 10^{-5} – 10^{-4} and, stripped of its second parameter, is the looser of the two on thirteen of the fourteen splits.

E. Where the threshold comes from

Measured rung by rung along the chain of inequalities that proves Theorem 1, at $L = 8$, the displacement of the threshold is 44% the pointwise step to a Hilbert-space distance, 38% the operator-norm step $\|R(z)\| \leq 1/\eta$, 11% the Feshbach/Minkowski split and 4.6% Cauchy–Schwarz in ω , whose total margin for tightening is a factor 1.05 typically and 3.29 at worst. The subspace adds one structural fact. For a subspace holding the whole probe, $\Lambda_S = 0$ requires $\text{span}(S) \supseteq \mathcal{K}(H, \phi)$, and the reflection $j \rightarrow -j$ about the seed site commutes with H and has ϕ as an eigenvector, so every vector of $\mathcal{K}$ vanishes on the reflection-fixed determinants of the opposite parity—4, 18, 48, 200 and 600 at $L = 6$ –14—while a reflection-projected power iteration (`z_suppK_sym.py`) finds every other determinant in the support: 296, 3902, 52 872, 731 608 and 10 305 696 at $L = 6$ –14. At full captured weight, therefore, no coordinate subspace short of all of them has zero leakage. That is a statement that $\Lambda_S \neq 0$, not about its size; the vacuity at the published fractions, all below full captured weight, is the measurement of Table III and not a deduction from it. Nor is the ranking the weak link: a Krylov-diagonal oracle ranking gives a *worse* certificate than the Born ranking at $L = 6$, $\eta = 0.18t$, at every fraction checked from 0.50 on. At full captured weight the leakage branch is of first order in the boundary coupling and the error of second, and the product $\Lambda_P \hat{\Lambda}_P / \eta^2$ is itself an a posteriori certificate of second order (Sec. S4); every published fraction lies below full captured weight, and whether a certificate of second order exists there is open. Finally, for the momentum-resolved probe of Fig. 3 the Krylov space lies inside one momentum block of exactly $\text{dim}(N+1)/L$ determinants, on which the certificate becomes non-vacuous at 48–50 of 50 determinants ($L = 6$) and 455–483 of 490 ($L = 8$) (`z_kblock.py`), a factor 5–7 fewer than the determinant basis requires; the gain is $1/L$ against an exponential, and whether the support fills the block at $L \geq 10$ is unmeasured.

VI. NO SHORTCUT TO THE BOUNDARY: AN OBSTRUCTION AND ITS WITNESS

Section III leaves one question open: can the a posteriori certificate be replaced by an a priori predictor computed from the state before any circuit runs? The natural candidates are the monotones of fermionic non-Gaussianity, efficiently measurable from two-point functions (Sec. S5 reviews them). None can play that role, for either term of Eq. (7): not for the static determinant support $|\mathcal{S}|_{\varepsilon_w}$ of the ground state that the shots must populate, and not for the leakage $\Lambda_S(\eta)/\eta$ of the coordinate subspace the sampler returned. What replaces the predictor is the measured map of Sec. V.

A. The collapse: one-body magic is a single 1-RDM invariant

We work with the fermionic AntiFlatness (FAF) [65],

$$\mathcal{F}_k = 2n_o - \frac{1}{2} \text{tr}[(\Gamma^\top \Gamma)^k] = 2n_o - \sum_{j=1}^{2n_o} \lambda_j^{2k}, \quad \Gamma_{mn} = -\frac{i}{2} \langle [\gamma_m, \gamma_n] \rangle, \quad (15)$$

with Γ the $4n_o \times 4n_o$ Majorana covariance matrix, $\{\lambda_j\} \in [0, 1]$ the moduli of its eigenvalue pairs $\pm i\lambda_j$, and $2n_o$ the number of spin-orbital *modes* (twice the spatial-orbital count), distinct from the electron number of the $(N \pm 1)$ sectors. Each $\mathcal{F}_k$ vanishes if and only if the state is Gaussian and is invariant under Gaussian (free-fermion) unitaries by construction [65]; the index k is a hierarchy of correlator orders, efficiently measurable from two-point functions [10, 65, 66].

On the number-conserving states of electronic structure the anomalous correlators $\langle c_i c_j \rangle$ vanish, so Γ is fixed by the one-particle reduced density matrix γ and its singular values are $\lambda_i = |2g_i - 1|$, with $g_i \in [0, 1]$ the natural spin-orbital occupations. The $k = 1$ member then reduces to a single 1-RDM invariant,

$$\boxed{\mathcal{F}_1 = \sum_i [1 - (2g_i - 1)^2] = 4 \sum_i g_i(1 - g_i) = 4 \text{tr}[\gamma(1 - \gamma)]}, \quad (16)$$

exact on any number-conserving state and verified state by state to 10^{-13} on the 38 molecular ground states of this work. For spin-restricted natural orbitals it equals $2N_u$, the effectively-unpaired-electron index [67, 68], and $\text{tr}[\gamma(1 - \gamma)]$ is the one-body-entanglement linear entropy [69]. The algebra is an immediate number-conserving specialization of the antiflatness of Ref. [65], noted independently for interacting-integrable eigenstates [70]; what we carry forward is its consequence. Being a function of the occupation spectrum alone, $\mathcal{F}_1$ is invariant under orbital rotations (the spin-polarized cases and the dictionary are in Sec. S5).

B. The geminal witness, and what an invariant cannot certify

The obstruction is not a continuity artifact. For the *raw* determinant support—the unthresholded Slater rank—no continuous functional of a fixed set of reduced density matrices can lower-bound the count at all, rank being coefficient-insensitive and not lower-semicontinuous, like the CP tensor rank [71]; that argument (Sec. S13) does *not* transfer to the operative, coefficient-*sensitive* $|\mathcal{S}|_{\varepsilon_w}$, which collapses to 1 along the same path as the functionals. What obstructs the operative quantities is a symmetry, exhibited by a family whose amplitudes are $\Theta(1)$ and therefore threshold-robust: an antisymmetrized product of K strongly orthogonal singlet geminals (perfect-pairing / generalized-valence-bond),

$$|\Psi_K\rangle = \bigotimes_{i=1}^K \left(\cos \theta \hat{b}_{i\uparrow}^\dagger \hat{b}_{i\downarrow}^\dagger + \sin \theta \hat{a}_{i\uparrow}^\dagger \hat{a}_{i\downarrow}^\dagger \right) |0\rangle, \quad (17)$$

on $2K$ spatial orbitals ($4K$ modes, $N = 2K$ electrons), each geminal occupying a disjoint bonding/antibonding pair (b_i, a_i) .

Theorem 3 (Geminal orbit witness). *At maximal pairing $\theta = \pi/4$ the family (17) satisfies, exactly,*

$$\mathcal{F}_1 = 2N_u = 4K, \quad \mathcal{F}_2 = 4K, \quad |\mathcal{S}|_{\text{pair}} = 2^K, \quad \chi = 2, \quad (18)$$

with $\mathcal{F}_{1,2}$ the order-1 and order-2 antifatness of Eq. (15), $|\mathcal{S}|_{\text{pair}}$ the determinant support in the pairing basis, and χ the minimal matrix-product-state bond dimension. One-body magic, higher-order magic and determinant support therefore all diverge with K while, in the localized (pairing) basis, the state remains a bond-dimension-2 tensor network—representable, contractible and Born-samplable in $O(K)$: magic of every order is unbounded at fixed $O(1)$ simulation cost.

The proof is four block-diagonal computations (Sec. S12), and all four values are reproduced by exact diagonalization for $K = 1, \dots, 4$ (Fig. S1). The orbit is the argument: $\mathcal{F}_k$ is invariant under number-conserving orbital rotations [65], so the whole orbit $\{U|\Psi_K\rangle\}$ carries the same $(\mathcal{F}_1, \mathcal{F}_2) = (4K, 4K)$ while the cost in the fixed computational basis runs from $O(K)$ to exponential. A Gaussian-invariant fermionic magic therefore neither lower-bounds the cost nor determines it. The same orbit reaches the second budget of Eq. (7), where an invariant upper bound on the leakage exists but is useless:

Proposition 1 (Fibers on which every orbital-rotation invariant gives a vacuous leakage certificate). *Let $\mathcal{U}_U$ be the Gaussian unitary implementing $U \in \text{U}(n_o)$, and let the same physical problem written in the rotated orbital basis be $(H_U, \Psi_U, \phi_U) = (\mathcal{U}_U^\dagger H \mathcal{U}_U, \mathcal{U}_U^\dagger \Psi, \mathcal{U}_U^\dagger \phi)$, the determinant basis—in which the device measures, and of which S is a coordinate subspace—held fixed. Along this orbit the spectrum of H , E_0 , $\|\phi\|^2$, every Lehmann weight and hence $A(\omega)$ pointwise are constant, as is every orbital-rotation invariant of the state, in particular $\mathcal{F}_1 = 4\text{tr}[\gamma(1-\gamma)]$ and all $\mathcal{F}_k$. $\Lambda_S(\eta)/\eta$ is not. An upper bound on $\Lambda_S(\eta)/\eta$ is available from invariants alone—combining $\Lambda_S \leq \|P_S H Q_S\| \|\phi_S\|$ with $\|Q_S H P_S\| = \|Q_S(H - c)P_S\| \leq \min_c \|H - c\|$ gives $\Lambda_S(\eta)/\eta \leq (W/2)\|\phi\|\sqrt{w_S}/\eta$, W the spectral width of the $(N \pm 1)$ sector—but it is vacuous by a factor of at least $W/2\eta$, which grows extensively with system size. That this is the best such bound can be seen at fixed invariants: any $B(\text{invariants}, |\mathcal{S}|, w_S)$ valid for $\Lambda_S(\eta)/\eta$ is constant on a fiber of fixed arguments and must therefore exceed the fiber maximum. In the geminal family APSG($\pi/4$) on $2K$ orbitals with $H = \varepsilon \hat{N} - g \sum_i A_i^\dagger A_i + \delta h$ we exhibit fibers ($K = 2, 3, 4$; $\delta = 0.01$; $\eta = 0.10$; $|\mathcal{S}| = 2^{K-1}$; spectrum, Lehmann weights and all $\mathcal{F}_k$ matched to 10^{-14} , w_S to 2.2×10^{-16}) whose endpoints carry $\hat{\Lambda}_S(\eta)/\eta = 0.237$ and $2.326, 0.323$ and $2.336, 0.391$ and 2.346 . Since the larger value exceeds $\tau(w_S) = (1 + w_S)/(1 + \sqrt{w_S}) - \sqrt{1 - w_S} = 0.795$, the resolution above which the leakage branch of Theorem 1 no longer improves on the trivial bound $\|\phi\|^2(1 + w_S)$, every valid invariant bound is vacuous on these fibers—while the subspace-aware certificate is non-vacuous at the aligned endpoint (0.856 against a trivial 1.962). The statement is one of non-determination and non-vacuity, not of non-existence: invariants do not fail to bound the leakage, they fail to bound it usefully.*

The fibers (Table S4), an exhaustive search over the 276 subspaces of that size at $K = 2$, and four limits of the exhibit—a matched pair at one angle, a margin over the trivial bound that shrinks from 2.29 to 1.69 between $K = 2$ and $K = 4$, a small-leakage endpoint realized only by the stabilizer of the state, and sector dimensions of at most 3920—are in Sec. S5.

The cost side: a bilateral obstruction whose sign is not stable. Unlike $\mathcal{F}_1$, the determinant support is basis dependent, and both directions of the decoupling are realized (Fig. S2): a free-fermion ground state has $\mathcal{F}_1 = 0$ and a site-basis support of 35, 336, 3496 and 37361 at $L = 4$ –10; along a Hubbard chain raising U raises $\mathcal{F}_1$ while lowering $|\mathcal{S}|_{\varepsilon_w}$; across the nineteen molecules the two rise together. Within each of the six (L, N) strata of the Hubbard sweep the rank correlation of $\mathcal{F}_1$ with $|\mathcal{S}|_{\varepsilon_w}$ is exactly -1 (stratified Kendall $\tau = -1.000$, exact one-sided permutation $p = 3.35 \times 10^{-13}$ against independent orderings within each stratum, a null that six strata sharing one mechanism do not satisfy), while it is $+1.000$ in the chain and ladder families and $+0.86$ across the molecules: an excellent rank predictor whose sign reverses between families, as an orbital invariant compared with a basis-dependent count must. The bond dimension χ , a rigorous floor on the raw support, tracks the thresholded one better (mean within-stratum $\rho = 0.948$), but it is itself basis dependent, measured rather than predicted, and blind to the leakage. The division between Gaussian-invariant and basis-dependent quantities is due to Tarabunga *et al.* [10] and to Leone and Bittel [11]; what the construction adds is a family on which every order of the invariant is maximal at $O(K)$ cost, and fibers on which the invariants are frozen while the certificate of Theorem 1 still separates. The stratified statistics, the molecular set, which does not discriminate, and the two families where $\mathcal{F}_1$ wins are in Secs. S5 and S14.

VII. THREE PRICES: SUPPORT, RESOLUTION, AND SHOTS

Three axes are priced here: the determinant support the sampler must populate, the resolution η at which the response is asked for, and the shot budget T that populates it. It is easy to price the first two and to quote the second as though it were a property of the model. It is not. This section states which of the three exponents belongs to the Hubbard chain and which belongs to the question we put to it, and prices the third, which is usually left unpriced.

One symbol, three counts. $|\mathcal{S}|_\eta$ here is *dynamical*: the number of determinants needed for the reconstruction of $A(\omega)$ to reach relative $L_1 < \text{thr}$ at resolution η and recursion depth n_ℓ , so it depends on all three (`scaling_lanczos_mf.py`, `stage_frac`: `thr = 0.05`, $\eta = 0.15 t$, $n_\ell = 150$). It is not the static ground-state support $|\mathcal{S}|_{\varepsilon_w}$ of Sec. VI, nor the returned set S of Theorem 1; the three coincide only by accident, and no statement here transfers to the other two.

A. The support exponent, and what does and does not survive a change of resolution

Across $L = 4\text{--}14$ (8 to 28 qubits, half-filled *periodic* rings at $U/t = 8$; the three largest sizes reached by matrix-free Lanczos/Haydock and validated against dense diagonalization at $L \leq 8$), the determinant support needed to reconstruct $A(\omega)$ to relative $L_1 < 0.05$ at $\eta = 0.15 t$ and $n_\ell = 150$ grows exponentially,

$$|\mathcal{S}|_{\eta=0.15} = 22, 246, 2180, 19183, 124407, 824503 \implies |\mathcal{S}|_\eta \sim e^{1.05 L} \quad (L = 4\text{--}14, R^2 = 0.998), \quad (19)$$

more slowly than the $(N+1)$ sector itself, $\dim = 24, 300, 3920, 52920, 731808, 10306296 \sim e^{1.30 L}$. Restricting the fit to $L = 4\text{--}12$, the five sizes of the replication grid, gives 1.082 at $R^2 = 0.998$, and the fit range is quoted with the exponent throughout because it matters at the second decimal. The fraction is the support divided by the sector dimension, exactly, so the fraction and support exponents differ by the constant $b_{\dim} = 1.291$ and move together, by 0.148 over the decade $\eta = 0.05 \rightarrow 0.50 t$ (Table S5): a seventh of the support exponent, three quarters of the fraction exponent. We therefore report the support, the absolute resource a device has to populate, and the fraction as the derived, resolution-dependent quantity that it is.

B. The sector fraction is a property of the question, not of the model

The headline these numbers offer—that the fraction needed for a fixed spectral accuracy falls from 0.92 at $L = 4$ to 0.08 at $L = 14$ —is tunable, and no headline of that form is claimed here: over the decade of η the fraction exponent moves by a factor 2.2, from -0.1266 to -0.2743 (Table S5), and even at fixed η the local logarithmic slopes between consecutive published points steepen by a factor 6.8 across five intervals. The reference is itself a depth-150 Lanczos computation whose drift between depths 100 and 200 at $L = 12$ is 0.93% at $\eta = 0.15 t$ and 3.0% at $\eta = 0.10 t$, so the fraction is reported as a declared window, $\eta \geq 0.15 t$ at $n_\ell = 150$, with the full grid and the finite-depth caveats in Sec. S6.

C. The third price: shots

Populating a subspace by measurement is a coupon-collector problem on the Born distribution of the time-evolved seed, so the budget exceeds the support it buys. Table IV reports that budget *measured*, by multinomial sampling of the declared protocol, at every size but the largest, in two independent runs that agree where they overlap to 0.2% and 3.2%; every budget is for *one* orbital seed and *one* branch.

Four statements are supported by the measured rows, and a fifth is not. Supported: the budget is two orders of magnitude larger than the support it buys, already at $L = 6$; the budget itself grows exponentially, 7.8×10^2 to 1.9×10^7 over $L = 4\text{--}12$, a log-linear fit over $L = 6\text{--}12$ giving $T \sim e^{1.12 L}$ at $R^2 = 0.996$; at $L = 8$ the full momentum-resolved observable costs $16 \times 2.7 \times 10^5 \approx 4 \times 10^6$ shots, which is a device experiment, while the $L = 14$ floor of $\sim 10^8$ for a *single* seed and branch is neither a laptop experiment nor, at present readout rates, a device one; and the overhead per determinant is a level and not a growth law—35, 99, 124, 161, 154 against the published support across $L = 4\text{--}12$, rising by a factor 4.4 over the first three sizes and flat within the seed spread over the last two. Not supported, and not written: an exponent for that overhead. Fitted on $L = 6\text{--}12$ it is 0.08 per site against the published support (95% interval $[-0.03, 0.19]$) and 0.054 ($[-0.066, 0.174]$) against the support each run populated; both intervals contain zero, so *these data do not establish that the shot budget grows with a larger exponent than the support*, only that it is larger in absolute terms at every size measured.

What the published curve is. The ranking that produced the published fractions is `argsort` applied to the Born distribution of the time-evolved state accumulated over the 19 snapshots of the protocol—the distribution the device would itself measure, in the limit of infinitely many shots. It is not a clairvoyant ordering by exact eigenvector amplitudes, and it is not a finite-shot reconstruction: the published curve is the $T \rightarrow \infty$ limit of the declared protocol. With finite shots the fraction needed rises, from 0.8200 to 0.8571 ± 0.0135 at $L = 6$ and from 0.5561 to 0.6275 ± 0.0031 at $L = 8$ (Table S6), so the published curve is a strict lower bound on the finite-shot cost. The penalty is budgetary, not

TABLE IV. **The third price, measured at every size but the largest.** T is the total shot budget, split evenly over the 19 snapshots of the protocol, for a single orbital seed and a single branch; it is the budget at which the finite-shot reconstruction itself first reaches relative $L_1 < 0.05$, and $\pm$ is the sample standard deviation across seeds, not the standard error of the mean. Run A is `sampled_honest.json` (8 seeds, $L \leq 8$); run B is `honest_sampling.json` (4 seeds, 6 at $L = 10$), and the two are separate runs with different seeds and different budget grids, listed separately at $L = 6$ and $L = 8$ so that their agreement can be read rather than asserted. At $L = 4$ run B is censored by its own cap of 10^8 shots per snapshot, the 24-state sector being degenerate for this purpose, and run A is quoted there. The last column divides T by the *published* support; dividing instead by the support each run actually populated gives 95, 107, 141 and 124 at $L = 6$ –12. Only $L = 14$ is a floor: the overhead held at its $L = 12$ value, which is a lower bound and not an estimate.

L	$\dim(N+1)$	$ \mathcal{S} _\eta$	T	$T/ \mathcal{S} _\eta$	status
4	24	22	$(7.8 \pm 2.0) \times 10^2$	35	measured, run A, 8 seeds
6	300	246	$(2.43 \pm 0.34) \times 10^4$	99	measured, run A, 8 seeds
6	300	246	$(2.44 \pm 0.23) \times 10^4$	99	measured, run B, 4 seeds
8	3920	2180	$(2.70 \pm 0.13) \times 10^5$	124	measured, run A, 8 seeds
8	3920	2180	$(2.61 \pm 0.08) \times 10^5$	120	measured, run B, 4 seeds
10	52920	19183	$(3.09 \pm 0.04) \times 10^6$	161	measured, run B, 6 seeds
12	731808	124407	$(1.91 \pm 0.01) \times 10^7$	154	measured, run B, 4 seeds
14	10306296	824503	$\gtrsim 1.3 \times 10^8$	154	floor, not measured

TABLE V. **The two collective scales, across four system sizes.** Charge: the Mott gap $\Delta = \mu^+ - \mu^-$ from exact ($N \pm 1$) ground-state energies (half-filled Hubbard ring, $U/t = 8$; particle-hole symmetry $\mu^+ + \mu^- = U$ verified to 10^{-8} at every size, ARPACK checked against dense `eigh` on every sector of dimension ≤ 5000). $\Delta_\infty = 4.67952t$ is the Lieb–Wu thermodynamic value [45], evaluated here from its exact integral representation to an absolute quadrature error of 2×10^{-9} and cross-checked against a second exact representation, the two agreeing to 10^{-13} . Spin: the lowest pole of $S^{zz}(q = \pi, \omega)$, a pole position of the tridiagonal recursion and therefore independent of the broadening used to display it. Right-hand block: the same quantity in the pure Heisenberg ring at the effective exchange $J = 4t^2/U = 0.5t$, a model with no charge degree of freedom at all. The residual column is quoted to four decimals so that it is the difference of the two columns to its left; at five it would not be.

L	charge		spin (Hubbard)		spin (Heisenberg, $J = 0.5t$)	
	Δ/t	$(\Delta - \Delta_\infty)/t$	Δ_s/t	$L \Delta_s/t$	Δ_{ST}/t	$L \Delta_{ST}/t$
6	5.35813	0.6786	0.34857	2.091	0.34237	2.054
8	5.18584	0.5063	0.23958	1.917	0.26134	2.091
10	5.02224	0.3427	0.19956	1.996	0.21162	2.116
12	4.96876	0.2892	0.16635	1.996	0.17792	2.135

structural: at $L = 8$ raising the budget by a factor 12.7 takes the ratio of the finite-shot error to the infinite-shot-ordering error from 1.96 ± 0.12 to 1.06 ± 0.01 . Finite-shot fractions are therefore reported only with their budget and seed count attached.

VIII. PHYSICAL VALIDATION: THE TWO GAPS SCALE IN OPPOSITE DIRECTIONS

The two collective channels of the half-filled Hubbard chain are the strongest physical check available to this method, because their answer is known exactly and independently of it. This section reports that check as a scaling statement across $L = 6, 8, 10, 12$ rather than as a picture at a single size, does not use three words that a single size cannot support, and separates what is a validation of the method from what is a result. On the same lattice and from the same primitive, the charge scale converges to a finite thermodynamic value while the spin scale closes as $\approx 2/L$, with an independent effective-Heisenberg control (Table V).

A. Charge converges to a finite value, spin closes as $1/L$

Charge. $\Delta(L)$ decreases monotonically and stays above the Lieb–Wu value $4.67952t$ at every size, the residual falling from $0.679t$ at $L = 6$ to $0.289t$ at $L = 12$, 6.2% of the thermodynamic value; $\Delta(L = 12) = 4.96876t$ is the

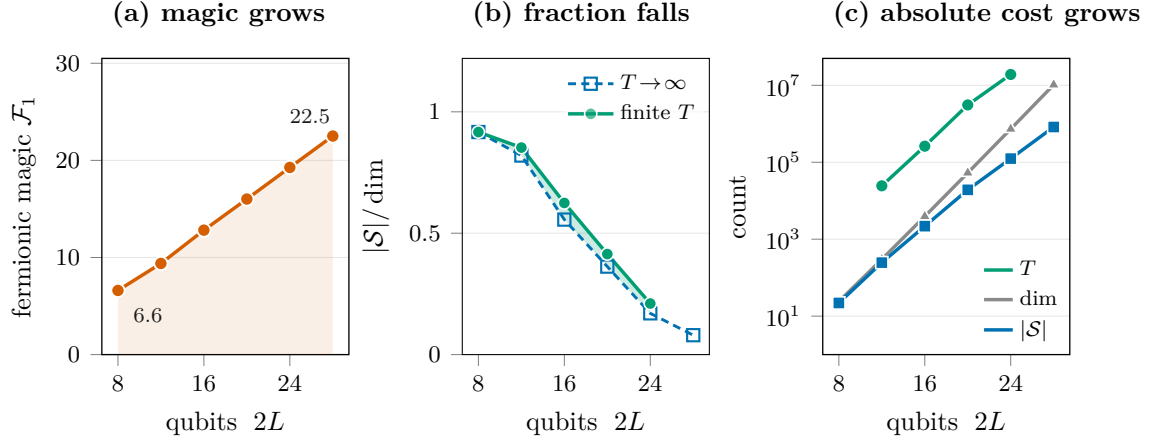


FIG. 5. **Three faces of the cost, measured with shots.** One-dimensional Hubbard ring, $U/t = 8$, half filling, periodic boundary conditions; the subspace is built in the $(N+1, S_z+1)$ sector seeded by $c_{0\uparrow}^\dagger|\Psi_0\rangle$ from the $K+1 = 19$ time snapshots of the declared protocol ($K = 18$, $\delta t = 0.5$), and a subspace is accepted when its Haydock reconstruction reaches relative- $L_1 < 0.05$ against the exact spectrum at broadening $\eta = 0.15t$. All fractions below are at that single η . **(a)** One-body fermionic magic of the ground state, 6.6074 at $L = 4$ to 22.5050 at $L = 14$ (periodic ring; the open-chain families elsewhere in this paper carry different values and must not be substituted here). **(b)** The required fraction of the sector, twice. Open squares, dashed: the infinite-shot limit of the same protocol — the subspace obtained by ranking the exact Born weights of the snapshots — 0.9167, 0.8200, 0.5561, 0.3625, 0.1700, 0.0800 for $L = 4$ –14, the published fractions quoted throughout this paper. Filled circles, solid: the same acceptance criterion reached by actually drawing shots, 0.9167 ± 0.0000 , 0.8527 ± 0.0096 , 0.6246 ± 0.0033 , 0.4141 ± 0.0012 , 0.2104 ± 0.0004 (mean $\pm$ s.d. over 4 independent shot seeds; 6 seeds at $L = 10$); the error bars are drawn and are smaller than the markers. Finite sampling needs +4.0%, +12.3%, +14.2% and +23.7% more of the sector than the infinite-shot limit at $L = 6, 8, 10, 12$: the penalty does not cancel, it grows with size, and the shaded strip is that difference. An independent 8-seed run of a separately written sampler reproduces 0.8571 ± 0.0135 ($L = 6$) and 0.6275 ± 0.0031 ($L = 8$), agreeing with the curve to within one standard deviation. No finite-shot point is shown at $L = 14$: its Born weights were not computed, and they are not extrapolated. **(c)** The absolute counts that panel (b) normalises away, plus the third one. Sector dimension $\dim$ (grey) and the support $|S|$ of the reported subspace (blue) both grow exponentially; the measured number of shots T that reaches the finite-shot operating point of panel (b) (green) is 2.4×10^4 , 2.6×10^5 , 3.1×10^6 , 1.9×10^7 at $L = 6, 8, 10, 12$ (s.d. over seeds 2.3×10^3 , 8.1×10^3 , 3.8×10^4 , 1.2×10^5). $L = 4$ carries no shot count: its 24-state sector is degenerate for this purpose and its budget is censored by the 10^8 -per-snapshot cap. Log-linear fits over the six sizes drawn in (c) (four for T , which has no $L = 4$ and no $L = 14$ point) give $\dim \sim e^{1.30L}$ ($R^2 = 0.9999$), $|S| \sim e^{1.05L}$ ($R^2 = 0.998$) and $T \sim e^{1.12L}$ ($R^2 = 0.996$, four sizes). Those exponents are fits to the points this panel plots, not to the five-size fits stored in `honest_sampling.json` (1.108 and 1.082). Over $L = 6$ –12 the exponent of T exceeds that of the populated support by 0.054 per site ($[-0.066, 0.174]$) and that of the plotted support by 0.08 ($[-0.03, 0.19]$): these data do *not* establish that the shot budget grows with a larger exponent than the support, only that it is larger in absolute terms at every size measured (Table IV). **Resolution dependence.** Over $\eta/t \in [0.05, 0.50]$ the exponent of $|S|$ moves from 1.165 to 1.017 and that of the fraction in panel (b) by a factor 2.2, from -0.1266 to -0.2743 (Table S5, $L = 4$ –12). In absolute value both move by the same 0.148: they are one exponent, and neither is stable.

Hubbard-band separation of Fig. 4. No extrapolation exponent is fitted: a straight line in $1/L$ undershoots Lieb–Wu by $0.12t$, which is evidence that the form is not $1/L$, not evidence about the limit.

Spin. The lowest pole of $S^{zz}(q = \pi, \omega)$ gives $L\Delta_s = 2.091, 1.917, 1.996, 1.996$: a $1/L$ closing with a coefficient of about 2 (mean 2.000, range 1.917–2.091), and a fit $L\Delta_s = d_0 + d_1L$ has a slope consistent with zero ($d_1 = -0.010 \pm 0.018$). A linear-in- $1/L$ intercept of Δ_s is -0.020 ± 0.020 , against 4.561 ± 0.044 for the charge gap; but the pure Heisenberg ring, whose gap is known to vanish, gives $0.0142 \pm 0.0014t$ at these sizes, so a vanishing intercept is not by itself evidence that a gap closes, and the flatness of $L\Delta_s$ is what carries the statement. That Heisenberg ring at $J = 4t^2/U = 0.5t$ —a different Hamiltonian, Hilbert space and code path—reproduces the $1/L$ law, $L\Delta_{ST} = 2.054$ –2.135, differing from the Hubbard value by 6.5% at $L = 12$ with a residual that changes sign: it confirms the scaling and the coefficient to within 10% and does not pin the coefficient (Sec. S7).

The two channels therefore behave *qualitatively oppositely* under a change of system size—one converging to a finite thermodynamic value, the other extrapolating to zero as $1/L$ —with a factor 29.9 between them at $L = 12$, and with $1/L$ intercepts that separate. That opposition, measured at four sizes with an independent control on the smaller scale, is the physical statement this work makes. It is a statement about pole positions, not about lineshapes, so it does not depend on the broadening at which either response is plotted.

B. What a single size at $\eta = 0.10t$ could not support

A spin-charge statement cannot be read off a lineshape at a single size, for a reason of resolution. At $L = 12$ the spin scale is $0.16635t$ against the $\eta = 0.10t$ used for Fig. 8: the gap is 1.66η , i.e. 0.83 of the Lorentzian full width at half maximum. At that resolution a spin response with a $0.17t$ gap and one with none are not distinguishable by eye in a single-size plot. The figure could *illustrate* the physics; it could not establish it. Three terms—“continuum”, “gapless” and “edge of the continuum”—are therefore not licensed by it, and none is used for any channel of this work at this size.

What establishes the physics instead is the four-size scaling of Table V, plotted in Fig. 6, and a pole census (Table S7): between $L = 6$ and $L = 12$ the sector grows by a factor 2440 while the largest per-momentum count of poles above 1% of the channel maximum grows from 3 to 7 in the addition branch, from 4 to 9 in the charge channel and from 2 to 3 in the spin channel, a discrete spectrum whose lines are broadened together rather than a dense one forming. The census, the sum rule as measured, and the size convention of the exact-diagonalization literature together with the dynamical-DMRG resolution criterion [72] this work does not meet at $L = 12$, are in Sec. S7.

C. Spin-charge separation is a check of the method, not a result

That the spin and charge responses of the one-dimensional Hubbard model live on separate energy scales has been known exactly since Lieb and Wu [45], is textbook [73], was seen in angle-resolved photoemission thirty years ago [74, 75], is already legible in the spectral weight of twelve-site clusters embedded by cluster perturbation theory [76], and has been computed by dynamical DMRG since 2007 [3] in all three of the momentum-resolved channels reported here—in that reference’s notation the charge factor $N(q, \omega)$ at $L = 120$, the spin factor $S(q, \omega)$ at $L = 100$ and the one-particle function $A(q, \omega)$ at $L = 90$, which are respectively our $S(q, \omega)$, our $S^{zz}(q, \omega)$ and our $A(k, \omega)$, for the extended Hubbard model of which ours is the $V = 0$ line; the same literature treats the Hubbard chain at $L = 48$ with $\eta = 0.1t$ by correction-vector DMRG, and reaches $L = 64$ in the Heisenberg chain [4]. We present it here as what it is: a check of the exact full-sector references, against which the sampled reconstructions are scored, against a known answer, with the des Cloizeaux–Pearson lower boundary [46] and the Müller upper bound [77] recovered as thermodynamic-limit references rather than as findings. No part of it is offered as new physics.

IX. HARDWARE, SCOPE, AND THE REGION WHERE THE METHOD HAS NO ADVANTAGE

A. The two hardware runs, and what “exact by coverage” means

Both device runs are preserved in full and are the entirety of the quantum hardware used in this work. The first, on the IBM Heron processor `ibm_fez`, sampled the $L = 6$ Hubbard chain at $U/t = 4$ with 12 qubits and 3.5×10^5 shots, and post-selection on particle number and S_z retained all 300 determinants of the $(N+1)$ sector, so the reconstructed A_{hw} equals the exact A *by coverage*: Rayleigh–Ritz on the whole sector is its diagonalization, and the vanishing error is an identity rather than a measurement of device fidelity (Fig. 9(a)). The circuits prepared a computational-basis $(N+1)$ determinant rather than $\hat{c}_p^\dagger |\Psi_0\rangle$ and applied $R_{xx}R_{yy}$ hopping without Jordan–Wigner strings, so they are not the primitive of Fig. 1. The post-selection, moreover, read each bitstring in reversed qubit order, which reflects the chain and exchanges the two spin species: under that reading none of the 3.5×10^5 shots of a noiseless simulation of the same seven circuits is retained, against all of them under the correct reading, so every one of the 300 retained determinants arose from a device error (`src/hw_bitorder_check.py`). The run therefore establishes only that the steps of the pipeline execute end to end on a processor, with a post-selection that was itself wrong, not that the device sampled the intended distribution, and nothing about accuracy at complete or incomplete coverage. The second, stretched N_2 at 24 qubits on `ibm_marrakesh`, reaches 0.59 ± 0.14 mHa of the exact active-space energy after recovery, below the 29.5 mHa of the noiseless simulation of the same circuit (Fig. 9(b)). The record does not store the recovered subspace sizes on which that comparison turns, so it cannot say whether device noise helps, although the mechanism is established elsewhere [78]. What is controlled is recovery against no recovery on the same corrupted bitstrings, and there the gain grows with the noise (Figs. S8 and S7). These runs are smaller than the spectral records of Refs. [5–7] and are not offered as a scale result; the full account, the job identifiers and the shot arithmetic are in Sec. S8.

The molecular suite. The nineteen-molecule suite and the dimensions of its sectors are in Sec. S15.1. Twelve of the nineteen reconstructions are exact by construction, because their sampled subspaces, of at most twelve determinants, already contain the probe’s Krylov space, so the suite cannot show that the reconstruction is efficient across chemistry. On the seven Hamiltonians with sectors of dimension 20 to 300 the same primitive reproduces the orbital-projected

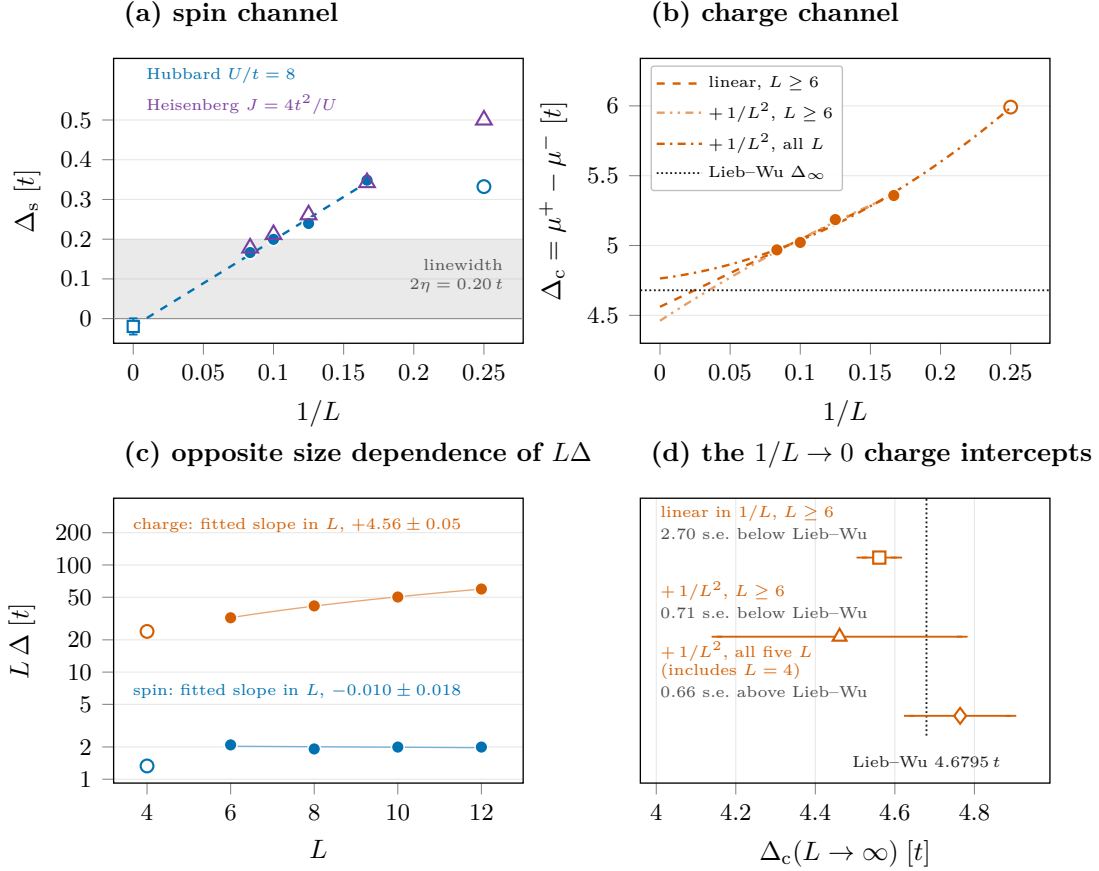


FIG. 6. The two gaps of the half-filled Hubbard ring scale in opposite directions. Periodic ring, $U/t = 8$, $L = 4, 6, 8, 10, 12$; in (a) and (b) the abscissa is $1/L$, so the sizes run $L = 12, 10, 8, 6, 4$ right to left, and open symbols are $L = 4$, which the $L \geq 6$ fits exclude. Every gap is an exact Lanczos pole, full reorthogonalisation, *no* broadening. **(a)** Spin gap, lowest pole of $S^{zz}(q = \pi, \omega)$: 0.34857, 0.23958, 0.19956, 0.16635 t at $L = 6, 8, 10, 12$ (0.33232 t at $L = 4$). Dashed: linear fit in $1/L$, $L \geq 6$ ($R^2 = 0.988$); the open square is its intercept $-0.0196 \pm 0.0204 t$, 0.96 s.e. from zero. Triangles: the pure Heisenberg ring $J = 4t^2/U = 0.50$, an independent control on a different code path, whose lowest $S^{zz}(q = \pi)$ pole reproduces its singlet-triplet gap to 8×10^{-15} . For $L \geq 6$ the two channels agree to within 8.3% (ratios 1.018, 0.917, 0.943, 0.935); at $L = 4$ they do not (0.665), which is why that size is excluded. Grey band: the linewidth actually used for the published spin figure, FWHM = $2\eta = 0.20 t$ at $\eta = 0.10 t$. The $L = 12$ gap lies inside it (0.16635 $t = 0.83$ FWHM), and broadened by this η it still carries 26.5% of its peak height at $\omega = 0$. One size alone therefore cannot separate a closing gap from no gap. **(b)** Charge gap $\Delta_c = \mu^+ - \mu^-$ from the $N \pm 1$ sectors: 5.3581, 5.1858, 5.0222, 4.9688 t at $L = 6, 8, 10, 12$ (5.9914 t at $L = 4$). At $L = 12$ that is 27.6 $\eta = 13.8$ FWHM for the $\eta = 0.18 t$ of the one-particle channel, 16.6 times further from $\omega = 0$ in units of its own linewidth than the spin gap. Three extrapolations are drawn and they disagree. The ordinate omits zero because the question here is which finite value the gap tends to, not whether it vanishes. **(c)** The same data with no extrapolation model, logarithmic ordinate: $L\Delta_s$ is flat over $L \geq 6$ (mean 2.0 t , the four values spanning 8% and not monotone, so no third figure is quoted; fitted slope in L -0.010 ± 0.018 , 0.57 s.e. from zero) while $L\Delta_c$ grows linearly ($+4.56 \pm 0.05$, 101 s.e.). The thin lines are those two linear fits, curved here only because the ordinate is logarithmic. **(d)** Those three intercepts on a scale where their standard errors are legible; on the scale of (b) the $\pm 0.0439 t$ bar is thinner than its marker. Linear, $L \geq 6$: $4.5609 \pm 0.0439 t$, undershooting the exact Lieb-Wu gap 4.6795 t by 0.1186 t , 2.70 s.e. Same protocol plus a $1/L^2$ term: $4.4611 \pm 0.3084 t$, undershooting by 0.2184 t (0.71 s.e., standard error 2.4 times larger). Only the $1/L^2$ fit over all five sizes brackets Lieb-Wu ($4.7641 \pm 0.1278 t$, 0.66 s.e. above), and it is the one fit here that includes $L = 4$, the size (a) and (b) mark anomalous. These sizes therefore fix the $L \rightarrow \infty$ charge gap only to about ± 0.1 – $0.3 t$: compatible with Lieb-Wu once a $1/L^2$ term is allowed, not evidence for it. What all three exclude is zero, the weakest by 14.5 s.e. and the linear one by 104 s.e. *Caveat, because it limits (a).* A vanishing intercept is not by itself the evidence: the Heisenberg control, whose gap is known to vanish in the thermodynamic limit, has a linear-in- $1/L$ intercept $0.0142 \pm 0.0014 t$, 10.0 s.e. above zero, and its own $L\Delta$ drifts +6.8% from $L = 4$ to $L = 12$, both from its logarithmic corrections. The model-free statement is (c).

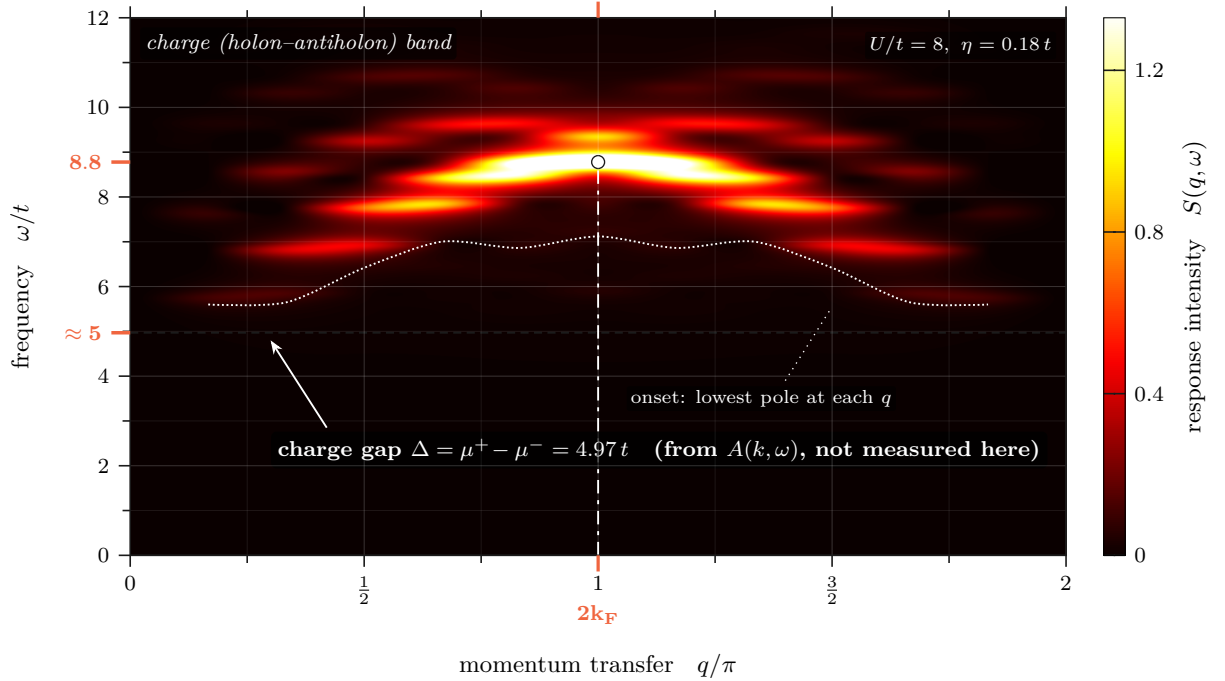


FIG. 7. **Dynamical structure factor: the density response.** $S(q, \omega)$ of the half-filled Hubbard chain ($U/t = 8$, $L = 12$, Lorentzian broadening $\eta = 0.18t$)—the density–density response, a number-conserving (neutral) excitation, computed by a Haydock continued fraction of depth $n_\ell = 150$ in the N -particle sector—a finite-depth quadrature of the spectral measure, not the Lehmann sum (Sec. S7). Colour is the *response intensity* $S(q, \omega)$ (units $1/t$): bright means the system responds strongly at that (q, ω) , dark means it does not. The weight forms the two-particle charge (holon–antiholon) band, whose dotted curve is the onset, the lowest pole resolved at each q , and sets the low-frequency scale of the band and traces its lower-boundary dispersion; the diffuse upper boundary is left unmarked. At $L = 12$ this band is six to nine discrete poles per momentum merged by the broadening, not a dense spectrum (Table S7). Two features are decisive. (i) The band is gapped on the scale of the charge gap: every resolved pole lies above $\Delta = \mu^+ - \mu^- = 4.97t$ (exact $(N \pm 1)$ ground-state energies), the lowest onset reached here being $5.60t$ at $q/\pi = 1/6$. The limit $q \rightarrow 0$ is inaccessible to this observable, $S(q=0, \omega>0)$ vanishing identically by density conservation, so Δ is drawn as a horizontal reference level imported from the single-particle channel and nothing is claimed about that limit. It is the *same* Mott gap that separates the Hubbard bands in the single-particle channel (Fig. 4): two independent dynamical observables, one correlation gap. (ii) The band peaks at $q = 2k_F = \pi$ at $\omega = 8.78t$ (coral tick, at the measured peak), within 10% of the bare Hubbard scale $U = 8t$ of a doubly-occupied site; the tick marks the peak, not U . The reflection $S(q) = S(2\pi - q)$ holds to $\sim 10^{-4}$ (Lanczos precision), and the analytic f-sum rule $\int \omega S(q, \omega) d\omega \propto (1 - \cos q)$ is recovered by the plotted (finite-window) grid to $\sim 0.3\%$. This is the collective observable the same bitstring-sampled primitive targets; as for $A(k, \omega)$, faithfully sampling the *full* response carries the sector-fraction cost of Sec. VII. This channel carries no shot budget: it is a full-sector reference (Table I).

$A(\omega)$ to a relative L_1 between 2.2×10^{-6} and 1.0×10^{-4} from 21% to 50% of the sector: a consistency check of the primitive outside the Hubbard family, not a scaling statement.

B. A region of no advantage, derived from our own resource statement

The cost tracker this paper puts forward, the bond dimension χ (Sec. VI), is by definition the cost parameter of a matrix-product representation of the same state: wherever it says the determinant support is small, a matrix-product state is small too. The instance is in the literature: the 7260 trajectories of a 120-qubit Fermi–Hubbard dynamics experiment [79] were reproduced by tensor contraction at bond dimension 32 [80], the pattern in which tensor-network, Pauli-propagation and matchgate methods have overtaken earlier hardware demonstrations [81–86]; on the ground-state energy the strong classical selectors already win on cost [87–89]; and in one dimension dynamical DMRG computed all three momentum-resolved channels reported here nineteen years ago [3]. Within the region this work measures—Hubbard rings at $L \leq 14$, nineteen active spaces, and a largest minimal bond dimension of 42 over the 74 deposited exact ground states—there is no computation performed here that a classical method cannot perform, and we make no

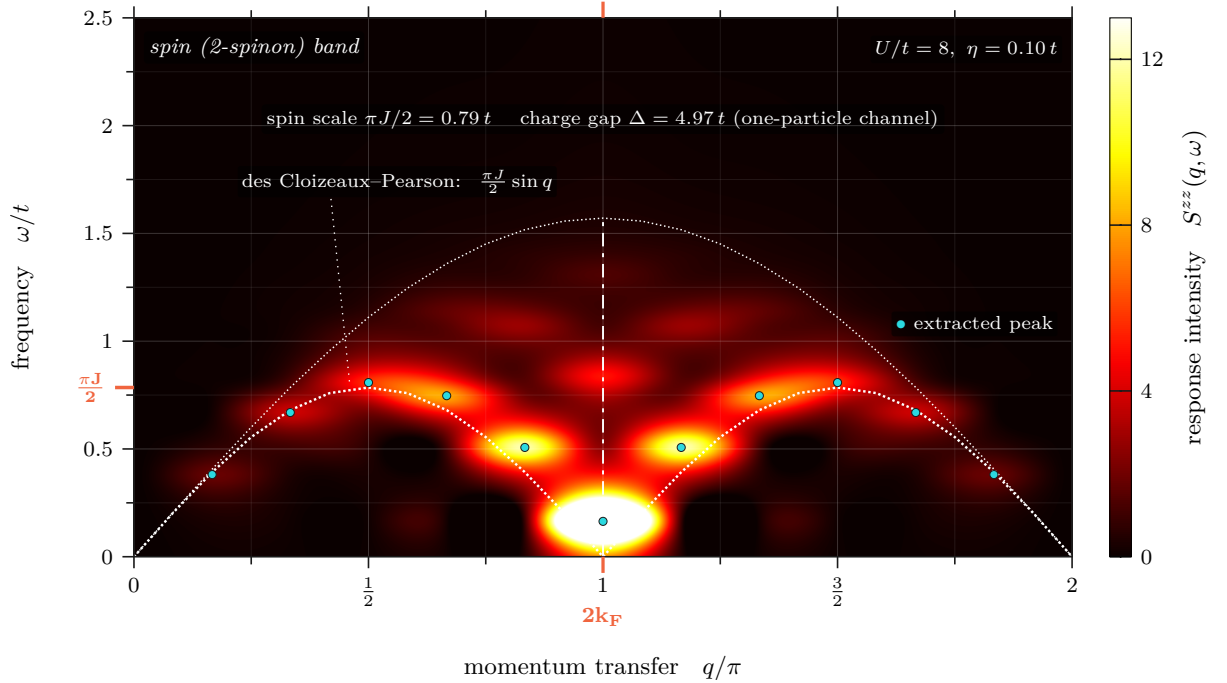


FIG. 8. **Spin structure factor: the low-frequency companion of the charge response.** The dynamical spin structure factor $S^{zz}(q, \omega)$ of the same half-filled Hubbard chain ($U/t = 8$, $L = 12$; here $\eta = 0.10 t$, finer because the spin scale is small), computed by a Haydock continued fraction of depth $n_\ell = 150$ from a spin seed $S_q^z|\psi_0\rangle$ in the N -particle sector; at this η that depth moves the reference by 3.0% between $n_\ell = 100$ and 200, so the lineshape is quoted with its depth and not as exact (Sec. S7). *Contrast Fig. 7:* the charge scale stays at the Mott gap $\Delta \approx 5 t$ at every size measured, while the spin scale is $0.16635 t$ at $L = 12$ and closes as $\approx 2/L$ (Table V). At this size the spin response is one dominant pole plus at most two satellites per momentum (Table S7), not a dense band, and what the figure shows is a set of broadened lines; whether the band touches $\omega = 0$ in the thermodynamic limit is not established here and is not claimed. This is spin-charge separation in the collective (two-particle) channel: charge and spin live in energetically separate worlds, the spin set by the superexchange $J = 4t^2/U = 0.5 t$, some $16\times$ smaller than U . The spectral weight piles up on the des Cloizeaux-Pearson [46] lower boundary $\frac{\pi J}{2}|\sin q|$ (dotted), bounded above by $\pi J|\sin \frac{q}{2}|$ [77]; the numerically-extracted peak dispersion (cyan markers) tracks this lower boundary to $\sim 3\%$; the antiferromagnetic soft mode at $q = \pi$ carries the dominant weight. The reflection $S^{zz}(q) = S^{zz}(2\pi - q)$ holds to double-precision round-off and $S^{zz}(q \rightarrow 0) \rightarrow 0$ (total- S^z conservation). The same bitstring-sampled primitive that targets $A(k, \omega)$ and $S(q, \omega)$ targets this collective response as well.

advantage claim at any size reached. Where that region would end is at a target whose minimal bond dimension is large in every ordering while the sampled support stays compact, a property of geometry and orbital connectivity that two-dimensional lattices and molecular orbital graphs would test and that nothing here measures (Sec. S8).

A dequantization test. On an under-sampled device head for N_2 (1 000 shots, ten paired seeds), a classical Epstein-Nesbet prior [90] applied to the same histogram cuts the energy error by 36%, from 38.22 to 24.38 mHa (paired $t_9 = 12.37$, all ten seeds), while a generative flow network [91] fused onto that prior moves it a further 1.80 mHa, a difference ten paired seeds do not resolve ($t_9 = 1.19$) and whose sign reverses between two five-seed halves; at 500 shots the fused generator is worse than the classical control (Sec. S16). On this instance the dominant determinants are supplied by the device sampling and completed by a classical prior rather than manufactured by a learned generator, and the largest improvement available to an under-sampled head costs no quantum resources at all.

C. Scope, declared

The momentum-resolved spectra are one-dimensional Hubbard rings at $L \leq 12$ with $\eta = 0.10\text{--}0.18 t$ (the support, certificate and shot measurements reach $L = 14$), which is the exact-diagonalization convention for this observable—Xu, Lu, Tohyama and Shao compute it at $L = 12$ and $\eta = 0.2 t$ at half filling, and at $L = 16$ at quarter filling, in the extended Peierls-Hubbard chain [92]—but the resolution criterion the dynamical-DMRG literature applies at those sizes is not met here, and Sec. VIII quotes that criterion in its original words and uses none of the three terms it would

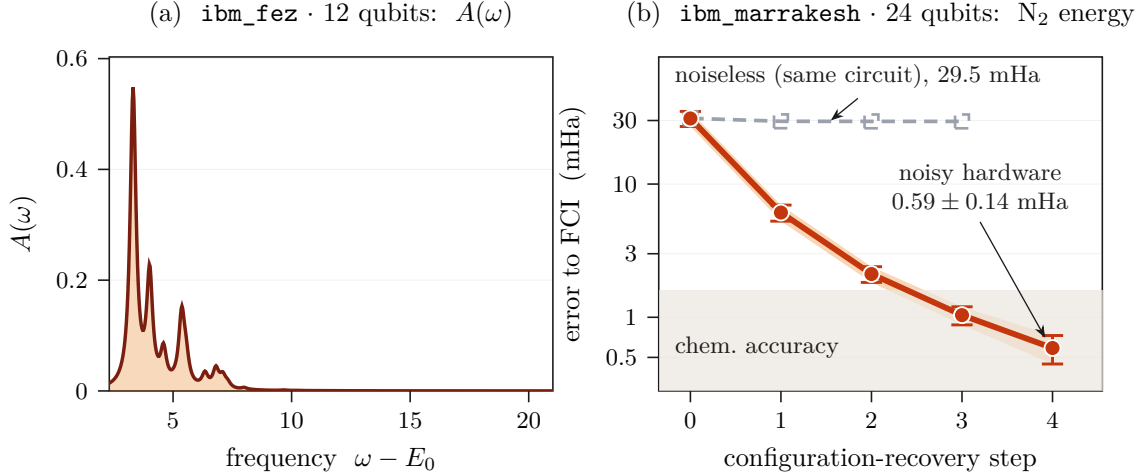


FIG. 9. **The complete quantum-hardware record of this work: two runs on IBM Heron processors.** (a) Single-particle spectral function $A(\omega)$ of the $L = 6$ Hubbard model at $U/t = 4$, $\eta = 0.15t$ (12 qubits; the classical validation of Fig. 2 is a different instance, $U/t = 8$), reconstructed from 350,000 computational-basis bitstrings—50,000 from each of seven circuits—sampled on `ibm_fez` (measurement and Pauli twirling and dynamical decoupling; no readout-error mitigation). The retained shots covered the complete $(N+1)$ sector ($|S| = 300/300$), so the reconstruction is exact by coverage; they were post-selected in reversed bit order, so every retained shot is a device error (Sec. IX A). The panel is an execution record, and neither a fidelity measurement nor a claim of advantage. (b) Ground-state energy error of stretched N_2 ($R = 2.0$ Å, CAS(10e,12o)/cc-pVDZ, 24 qubits) versus self-consistent configuration-recovery step, on `ibm_marrakesh` (10,000 shots, double-factorized Trotter circuit). The noisy device (solid) reaches 0.59 ± 0.14 mHa of the exact FCI value—well below chemical accuracy—whereas the noiseless statevector simulation of the *same* shallow circuit (dashed) is stranded at 29.5 mHa. That contrast is not read here as device noise *helping*: the mechanism such a reading needs is decided by the two recovered subspace sizes, and the record stores neither (Sec. IX A). The solid curve is the mean over 8 recovery seeds on the cached device counts; the shaded envelope and the capped whiskers at every step both show the ± 1 s.d. spread across those 8 seeds (widest at step 0, 31.1 ± 4.0 mHa, where it starts together with the noiseless curve, and tapering as recovery converges to 0.59 ± 0.14 mHa; best seed 0.40 mHa). The systematic version across nineteen molecules under a controlled noise model is Fig. S8. These two runs are the entirety of the quantum hardware used here; every other result in this work is an exact classical simulation.

not license, rather than enlarge η . The published subspace curves are the infinite-shot limit of the declared ranking protocol, and finite-shot fractions are reported only where Table S6 attaches a measured budget and a seed count. The certificate of Theorem 1 is vacuous at every operating fraction of the resource scan (Fig. 3, at 85%, is the certified exception), by factors of 1.66 to 8.61 at the five sizes measured, so the reconstructions at those fractions are at present uncertified and the non-vacuity threshold is given as a bracket rather than as a number (Sec. V). The device record is the two runs of Sec. IX A, at 12 and 24 qubits, the first exact by coverage; the chemistry record is Table S11. No result in this work is offered as evidence of an advantage over a classical method, at any size reached here.

What the work does claim is four things, each with its measurement attached: one sampling primitive that targets four frequency-resolved channels, of which the single-particle ones are reconstructed here from ranked or sampled subspaces and the two neutral ones are full-sector references (Table I); a two-sided a posteriori bound on the error of the reconstruction, with its non-vacuity threshold measured at five sizes up to a sector of dimension 10 306 296 and its crossing calibrated to a true relative error of $\approx 3 \times 10^{-3}$ within a factor of three (Sec. V); an obstruction theorem showing that no orbital-rotation invariant supplies a cheaper a priori predictor of that bound (Sec. VI); and three measured prices—support, resolution and shots—of which the third is measured here (Sec. VII).

D. Data availability statement

The data that underlie the figures and the quoted numbers of this work are openly available in the reproduction repository [93] at <https://github.com/nicolasbonilla/dynamical-spectral-functions-sqd>, archived on Zenodo under DOI 10.5281/zenodo.22967220, the code under the MIT licence and the manuscript, the deposited datasets and the reproduction documentation under CC BY 4.0. Every figure’s plotted values are deposited—as a JSON table in `data/` or as a plain-text table in `paper/figs/`—and `docs/FIGURE_PROVENANCE.md` names the file that is authoritative for each; the one exception is Fig. 1, a circuit schematic that plots no data. The classical data—the

exact-diagonalization spectral functions of the one-dimensional Hubbard model, the molecular suite and the resource statistics—are additionally *regenerable*: the computation scripts that produced them are deposited in `src/`, and `docs/REPRODUCE.md` gives the reproduction commands together with a measured table of what a clean clone does and does not reach. An automated guardian (`python src/verify.py`) recomputes the underlying physics by exact diagonalization and checks the deposited artefacts against it—874 checks over 9 891 numeric assertions, exiting non-zero on any discrepancy. It covers the deposited data files and the numbers printed in the figure sources, in the tables and in the abstract; it is not a proof that every sentence of this manuscript is verified, and five items are printed by name on every run: four known open defects, and one declared gap in its own coverage, the source of Fig. S5.

The quantum-hardware results are deposited as the arrays that enter the figures, together with the backend name, the shot count and the IBM Quantum job identifier for each of the two runs (`data/heron_spectral.json`, `data/hw_lucj_n2_result.json`). *The raw measured bitstrings are not part of the deposit*: the notebooks in `notebooks/` retrieve the counts from the IBM Quantum service by job identifier, which requires an account and a job that is still retrievable, so the classical post-processing of the raw counts cannot be re-run from the deposited files alone. The hardware runs themselves cannot be regenerated in any case—the jobs are closed, and a device executed today has a different calibration, so new counts would be new data rather than a reproduction. The dequantization control of Sec. S16 is regenerated seed by seed by `src/gflow_dequant.py`, which needs `pyscf` and `torch` and writes `data/gflow.json`. The $L = 14$ ground-state checkpoint (94 MB), on which the largest vacuity factor and the $L = 14$ support count rest, is deposited in `ckpt/` with its SHA-256 checksums. Known gaps between the deposited code and the deposited artefacts are enumerated in `docs/KNOWN_DISCREPANCIES.md`, and those still open are these: the standalone $\text{\LaTeX}$ sources of the six figures that enter as PDF are deposited, but the rasters embedded in three of them, Figs. 4, 7 and 8, are deposited as rendered and cannot be regenerated from the deposit; the deposited generator of Fig. 5 emits a two-panel version of it, the third panel having been added by hand, and the captions of Figs. 3 and 6 were edited by hand after generation, so none of these three is reproduced as printed by its generator; two deposited computation scripts have no deposited output; and parts of the pipeline require `pyscf` or an IBM Quantum account.

X. CONCLUSION

For a spectral function reconstructed by Rayleigh–Ritz on any subspace, Theorem 1 bounds the L_1 error from below by the Born weight the subspace failed to capture and from above by that weight plus the Hamiltonian coupling that leaks across the subspace boundary at the resolution asked for, a quantity computed from the Ritz pairs without the answer. The captured weight alone cannot replace it: an analytic counterexample and a counting argument rule out any bound indexed on it (Sec. IIID). What a Krylov-adequate subspace does guarantee is moment exactness through order $2K + 1$ and no further (Theorem 2). And no orbital-rotation invariant bounds the boundary *usefully* in advance: one-body fermionic magic is constant along an orbit on which the determinant support runs from $O(K)$ to 2^K at bond dimension 2, and on fibers on which the subspace-aware certificate still separates (Theorem 3, Proposition 1), the general invariant/basis-dependent division being due to Tarabunga *et al.* [10].

Evaluated at five system sizes up to a sector of dimension 10 306 296, the certificate is vacuous at every operating fraction of the resource scan, by factors of 1.66 to 8.61, converged in Krylov depth at $L = 12$; only Fig. 3, at 85% of its sector, is certified, at 350–900 times its true error. Two results survive that measurement. The crossing of the trivial bound occurs at a true relative L_1 error between 1.2×10^{-3} and 7.3×10^{-3} over sixteen (L, η) cells, a drift, not scatter, ordered in both arguments and conservative in L , computed without the answer—with the limit that a two-constant rule in $1 - w_S$, fitted out of sample, localizes it more tightly on eleven of the fourteen splits, so we claim no superiority for the certificate as a tracker of the error against the captured weight. And in the determinant basis a device measures in, the coordinate support of the Krylov space $\mathcal{K}(H, \phi)$ is every determinant the reflection about the seed site allows, measured at $L = 6$ to 14, so at full captured weight a subspace holding the whole probe leaks unless it contains all of them (Sec. VE); the vacuity at the operating fractions, all below full captured weight, is measured and not deduced from that fact. The method is priced in support, resolution and shots (Sec. VII), and the full-sector references reproduce the opposite finite-size scaling of the charge and spin gaps, the spin gap closing as $\approx 2/L$ with an independent effective-Heisenberg control (Table V).

Three limits apply. *First*, the certificate does not certify the reconstructions at the operating fractions of the resource scan: at full captured weight a certificate of second order in the boundary coupling follows from the same identity, but below it, where those fractions lie, one is open. The certificate also refers both spectra to the exact ground-state energy (H1); in a shot-based pipeline E_0 is itself estimated on a sampled subspace, and the shift of $6.7 \times 10^{-4} t$ that 60% of the N sector gives at $L = 8$ leaves the bound almost unchanged, +0.20%, while multiplying the measured error by 16 (Sec. S2). *Second*, the largest momentum-resolved spectrum here is $L = 12$, by exact diagonalization, whereas dynamical DMRG computed the same three channels at 90 to 120 sites in 2007 [3], correction vectors reach $L = 48$ in the Hubbard chain [4], and the classical toolbox is broader still [94–99]; one dimension is where tensor networks are

near-optimal [100, 101], and nothing in this work competes with that standard. *Third*, the certified regime costs more shots than the method saves: moving from the operating fraction to the non-vacuous band multiplies the shots per snapshot needed for the marginal configuration by 7.5 to 10 at $L = 8$ and by 20 to 35 at $L = 12$, where diagonalizing the whole sector is already the cheaper option.

Whether a certified regime exists at low sector fraction is now a well-posed measurement, and the calibrated crossing is the instrument for making it. Whether it exists at all turns on two open questions: whether a certificate of second order in the boundary coupling can be evaluated below full captured weight without the answer, and whether some single-particle basis gives $\mathcal{K}(H, \phi)$ a coordinate support that is a small fraction of the sector. What this work provides toward both is a computable two-sided bound, a measured map of where it is informative and at what cost, and the structural reason it is not informative at the fractions sampling reaches. The proofs, the complete tables and the extended discussions on which these statements rest are in the Supplemental Material [102].

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AI tools were used in the preparation of this manuscript; the author takes full responsibility for its content.

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